

An Internship Report on
Crypto Portfolio Manager
&
Infosys Springboard

BY

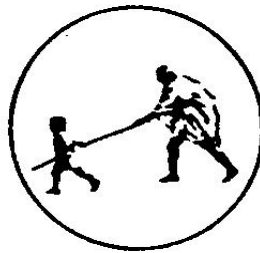
Moizuddin Siddiqui

Under the Guidance

of

Ms. K. P. Pople

(Department of Computer Science and Engineering)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Mahatma Gandhi Mission's College of Engineering, Nanded (M.S.)

Academic Year 2025-26

An Internship Report on
“Crypto Portfolio Manager”
&
“Infosys Springboard”

Submitted to
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE

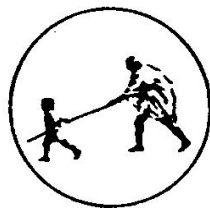
for fulfillment of the requirement for the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

By
Moizuddin Siddiqui

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of

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)

Academic Year 2025-26

Certificate



This is to certify that the Internship entitled

“Crypto Portfolio Manager”

*being submitted by **Mr. Moizuddin Siddiqui** to the Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by him under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

Ms. K. P. Pople

Project Guide

Dr. A. M. Rajurkar

H.O.D

Computer Science & Engineering

Dr. G. S. Lathkar

Director

MGM's College of Engg., Nanded

INFOSYS VIRTUAL INTERNSHIP 6.0 OFFER LETTER



Project Onboarding | Batch-13 | Infosys Springboard Virtual Internship 6.0

1 message

Internship Coordinator <coordinatorinternship02@gmail.com>

Sat, 31 Jan 2026 at 10:36 pm

Cc: springboardmentor576@gmail.com

Bcc: s22_siddiqui_moizuddin@mgmec.ac.in

Dear Intern,

Welcome to the Infosys Springboard Virtual Internship Program 6.0!

This mail is to inform you that you are onboarded into **Batch-13** under the Infosys Virtual Internship Program. We are excited to have you join us as an intern.

Your assigned project is: **Crypto Portfolio Manager - Group-2**

Classes for your project will be starting by next week!

We'll be sharing more information about the virtual internship soon, including the class schedule and meeting link. Please check your email regularly, including the spam folder, as all updates will be sent to your springboard registered email address.

We look forward to a successful internship experience with you!

Best regards,
Internship Coordinator

INFOSYS VIRTUAL INTERNSHIP 6.0 COMPLETION LETTER



CERTIFICATE OF COMPLETION

presented to

Moizuddin Siddiqui

for actively participating and completing the mandatory assignment related to

Internship 6.0 (B 13): Crypto Portfolio Manager

conducted from February 5, 2026 to April 3, 2026



Issued on: Monday, June 1, 2026
To verify, scan the QR code at <https://verify.onwingspan.com>

Satheesha B.N.

Satheesha B. Nanjappa
Senior Vice President and Head
Education, Training and Assessment
Infosys Limited

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I would like to express my deepest gratitude to my project guide, **Ms. K. P. Pople**, for her invaluable support, guidance, and encouragement throughout this project. Her profound knowledge and expertise have been instrumental in the successful completion of this work. Her patience and willingness to assist me at every step have greatly enriched my learning experience. Her constructive feedback and insightful suggestions have not only helped me overcome challenges but also motivated me to strive for excellence.

I gladly take this opportunity to thank **Dr. A. M. Rajurkar** (Head of Computer Science & Engineering, MGM's College of Engineering, Nanded). I am heartily thankful to **Dr. G. S. Lathkar** (Director, MGM's College of Engineering, Nanded) for providing facilities during the progress of the project and also for her kind help, guidance and inspiration. Last but not least, I am also thankful to all those who helped, directly or indirectly, develop this project and complete it successfully.

With Deep Reverence,

Moizuddin Siddiqui

[B.Tech CSE-A]

ABSTRACT

This report presents the development and implementation of the Python Crypto Investment Manager, created as part of the Infosys Springboard Virtual Internship Program 6.0 under the project *Crypto Portfolio Manager – Group 2*. The project focuses on building a smart and efficient crypto investment management system using Python to help users analyze, manage, and optimize cryptocurrency investments with risk-aware strategies.

During the project, various modules were designed and implemented to support investment planning, portfolio balancing, risk analysis, and automated reporting. The system uses Python's mathematical and analytical capabilities to calculate optimal asset allocation strategies and improve investment decision-making. Parallel processing techniques were incorporated to perform multiple risk checks and predictive operations simultaneously, improving the speed and efficiency of the application.

Through this project, practical exposure was gained in Python programming, concurrent task execution, data handling, database integration, automation, and financial analysis concepts. The development process also strengthened problem-solving, debugging, and logical thinking skills while demonstrating the importance of scalable and maintainable software design. Overall, this project bridges academic knowledge with real-world fintech and cryptocurrency portfolio management practices, preparing for future opportunities in software development, financial technology, and data-driven applications.

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chapter 1

INTRODUCTION

Internships play an important role in bridging the gap between academic learning and real-world industry experience. They provide students with practical exposure to professional environments, enabling them to apply theoretical concepts to real-time projects while developing technical, analytical, and communication skills. The Infosys Springboard Virtual Internship Program 6.0 offered an excellent opportunity to gain hands-on experience in software development and financial technology through industry-oriented projects. As part of this internship program, I was assigned to the project titled “Crypto Portfolio Manager” under the Crypto Portfolio Manager Group 2. The project focused on developing a smart cryptocurrency investment management system using Python. The main objective of the project was to design a platform capable of analyzing crypto investments, managing portfolio diversification, predicting risks, generating reports, and providing automated alerts to support better investment decisions.

The internship provided exposure to important concepts such as Python programming, mathematical analysis, concurrent task execution, database management, CSV report generation, and email automation. The project also involved implementing rule-based portfolio optimization and risk management techniques to help users maintain balanced and efficient crypto investments under varying market conditions. Throughout the internship, different modules of the project were designed and implemented systematically, including the Investment Mix Calculator, Risk Checker and Predictor, Report and File Saver, and Spreading Rule Setter. These modules collectively contributed to building an efficient portfolio management system capable of handling real-time investment analysis and reporting.

Apart from technical learning, the internship also enhanced problem-solving abilities, teamwork, debugging skills, project planning, and understanding of software development practices. Working on this project helped in understanding the practical challenges involved in developing fintech applications and strengthened knowledge of secure and maintainable coding practices. Overall this internship served as a valuable learning experience that combined academic concepts with practical implementation, preparing for future opportunities in software development, financial technology, and data-driven applications.

1.1 Internship Overview

An internship is one of the most important phases in an engineering student's academic journey, as it helps bridge the gap between theoretical knowledge and practical industry experience. It provides students with an opportunity to understand real-world work environments, modern technologies, project development workflows, teamwork, and professional responsibilities. Through internships, students gain practical exposure that enhances both technical skills and professional confidence, which cannot be fully achieved through classroom learning alone. As part of my final semester of B.Tech in Computer Science and Engineering, I successfully completed the Infosys Springboard Virtual Internship Program, where I worked on the development of the AI Powered Crypto Portfolio Manager project.

The internship provided me with valuable practical exposure to software development, financial analytics, AI-based systems, dashboard development, and real-time data visualization technologies. It allowed me to apply my academic knowledge to a real-world project environment and improve my understanding of modern application development practices. The internship followed a structured learning and development approach that combined technical training with practical project implementation. During the initial phase, I learned and explored various technologies such as Python, Streamlit, Pandas, NumPy, Plotly, AI-based forecasting techniques, and cloud deployment methods. As the internship progressed, I applied these technologies to develop different modules of the CRYPTOPORT platform.

Working on the project helped me improve my problem-solving ability, analytical thinking, coding skills, and understanding of real-time fintech applications. The internship also provided hands-on experience in designing interactive dashboards, integrating real-time cryptocurrency data, implementing forecasting systems, and developing user-friendly analytical interfaces. In addition, it enhanced my understanding of project management, testing, debugging, optimization, and cloud-based application deployment. The internship was a highly valuable learning experience that strengthened both my technical and professional skills. It gave me practical exposure to real-world software development workflows and helped me gain confidence in working with modern technologies related to financial analytics, artificial intelligence, and web-based application development. I applied these technologies to develop different modules of the CRYPTOPORT platform.

1.2 Company Profile

Infosys Springboard is a digital learning and virtual internship platform launched by Infosys with the objective of empowering students and learners through free access to industry-relevant education, technical training, and hands-on project experience. The platform provides learning resources in emerging technologies such as Artificial Intelligence, Cybersecurity, Cloud Computing, Data Science, Web Development, Python Programming, and Data Analytics. Cloud Computing, Data Science, Web Development, Python Programming, and Data Analytics. The internship program encourages innovation, technical problem-solving, teamwork, and professional



Fig 1.1: Company Logo

Fig 1.1 shows the official logo of Infosys Springboard, which represents innovation, learning, and digital transformation. The logo symbolizes the organization's commitment toward providing quality education, industry-oriented training, and practical learning opportunities to students through virtual internship programs and skill development initiatives. During this internship, Infosys Springboard provided the necessary platform, mentorship, and learning resources required for the successful development of the project “Crypto Portfolio Manager.”

The Infosys Springboard Virtual Internship Program is designed to bridge the gap between academic learning and industry requirements by providing students with practical exposure to real-world projects. The internship program encourages innovation, technical problem-solving, teamwork, and professional skill development through mentor-guided project implementation. As part of the Infosys Springboard Virtual Internship Program 6.0, the assigned project was “Crypto Portfolio Manager – Group 2”. with technologies related to financial analytics, and web-based application development. The project involved the development of an AI-powered cryptocurrency portfolio management system named CRYPTOPORT. The platform was developed to help users analyze cryptocurrency markets, monitor portfolio performance, evaluate investment risks, and visualize market trends using interactive dashboards.

The internship provided valuable exposure to software development practices, financial technology concepts, data visualization techniques, and AI-based analytical systems. It also improved practical knowledge in Python programming, Streamlit application development, portfolio analysis, forecasting methods, and interactive user interface design. Throughout the internship, hands-on experience was gained in designing and developing a professional web-based cryptocurrency management application with real-time analytical and visualization features.

1.3 Training and Skill Development

An important part of my internship was the technical learning and skill development process carried out during the Infosys Springboard Virtual Internship Program. This phase was designed to strengthen both theoretical understanding and practical implementation skills required for developing real-world software applications. The training focused mainly on Python programming, data analysis, dashboard development, AI-based analytics, and modern application development technologies used in the CRYPTOPORT project. During the internship, I improved my understanding of Python programming, which served as the core technology for the development of the AI Powered Crypto Portfolio Manager application.

I learned how to use Python for financial data processing, cryptocurrency market analytics also helped me gain practical experience with important Python libraries such as Pandas, NumPy, Plotly, and Streamlit. I developed hands-on experience in data analysis and visualization techniques using Pandas and NumPy for processing cryptocurrency datasets and performing financial computations. Using Plotly and Streamlit, I learned how to create interactive dashboards, graphical visualizations, forecasting charts, portfolio analytics dashboards, and real-time market monitoring systems.

These technologies helped me understand how modern analytical applications are designed and implemented in real-world environments. Another important learning area during the internship was AI-based forecasting and analytical modeling. I explored how historical cryptocurrency market data could be analyzed to generate predictive insights, portfolio risk analysis, and Buy, Hold, and Sell trading recommendations. This improved my understanding of financial analytics, predictive systems, and data-is, forecasting, portfolio management, and analytical calculations. The interndriven decision-making techniques.

The internship also provided practical exposure to cloud deployment, dashboard optimization, debugging, testing, and version control management using Git and GitHub. I learned how to organize source code, manage repositories, track project updates, and maintain proper development workflows throughout the project lifecycle. In addition to technical learning, the internship helped improve my problem-solving skills, logical thinking, analytical abilities, and confidence in handling real-world software development tasks. Working on the CRYPTOPORT project gave me valuable experience in developing a complete fintech-based analytical application with real-time visualization, AI integration, and interactive dashboard functionalities. Overall, the internship played an important role in strengthening both my technical knowledge and professional development skills.

1.4 Projects and Work Experience

During the Infosys Springboard Virtual Internship Program, the major project undertaken was “Crypto Portfolio Manager.” The project was developed as a web-based cryptocurrency portfolio management and analysis platform aimed at helping users monitor investments, analyze market trends, evaluate risks, and make better financial decisions using AI-driven insights and real-time analytics. The project was developed using Python and the Streamlit framework for creating an interactive and responsive user interface. Various modules were implemented to provide advanced portfolio management features such as cryptocurrency market tracking, portfolio analysis, AI-based trading signals, forecasting, and risk management. Real-time visualization techniques and interactive charts were integrated to enhance the overall user experience and provide meaningful financial insights.

Apart from technical skills, the internship contributed toward the development of problem-solving ability, logical thinking, teamwork, project planning, and debugging skills. Experience was also gained in managing project modules systematically, testing application functionality, and improving interface design for professional-level application development. The internship played an important role in strengthening both technical and professional competencies while providing practical exposure to real-world software development and fintech-based application implementation. Overall, the internship played an important role in strengthening both my technical knowledge and professional development skills. The internship played an important role in strengthening both technical and professional competencies.



Fig 1.2: CRYPTOPORT Project Dashboard

Fig 1.2 illustrates the main dashboard interface of the crypto portfolio manager developed during the internship project. The dashboard provides users with real-time cryptocurrency market information, portfolio insights, AI-generated trading signals, forecasting charts, and risk analysis tools in an interactive and user-friendly environment. The interface was designed using the Streamlit framework to ensure smooth navigation, professional visualization, and efficient access to financial analytics. The figure demonstrates the practical implementation of modern web technologies, data visualization techniques, and AI-based analytical features integrated throughout the project development process.

In addition to development work, I participated in code reviews, collaborated with team members through GitHub, and followed Agile practices such as daily stand-ups and sprint planning. This provided me with valuable insights into industry-standard project management practices. I also contributed to backend optimization efforts, identifying opportunities to improve API performance and refactor legacy code for better maintainability.

1.5 Learning Outcomes

The Infosys Springboard Virtual Internship Program provided valuable practical exposure to real-world software development and significantly enhanced both technical and professional skills. Through the development of the “Crypto Portfolio Manager” project, a deeper understanding of modern web application development, financial analytics, and AI-assisted portfolio management systems was achieved. During the internship, strong programming skills were developed in Python along with

practical experience in building interactive web applications using the Streamlit framework. Knowledge related to cryptocurrency market analysis, portfolio management, forecasting techniques, risk assessment, and data visualization was improved through hands-on implementation of multiple analytical modules.

The internship helped in improving practical skills in dashboard development, chart visualization, portfolio tracking, and implementation of AI-based trading signals. Experience was gained in handling financial datasets, presenting analytical insights, and designing professional user interfaces for enhanced user experience. Exposure to cloud deployment and web application hosting further strengthened practical software development knowledge. In addition to technical growth, the internship improved problem-solving ability, debugging skills, logical thinking, and project management capabilities. Working on a complete end-to-end project enhanced confidence in handling real-world development challenges and understanding the complete software development lifecycle including planning, implementation, testing, optimization, and deployment.

1.6 Organization of the Report

This report is divided into five chapters, with each chapter covering a different part of the internship experience and the project development completed during the Infosys Springboard Virtual Internship Program. It includes details about the organization, the learning process, the technologies used, the project implementation, and the overall outcomes and knowledge gained throughout the internship.

Chapter 1: Introduction – This chapter provides an overview of the internship program, including the internship objectives, company profile, training activities, project overview, learning outcomes, and overall internship experience. It also introduces the project “Crypto Portfolio Manager” and explains its purpose and significance in the field of cryptocurrency portfolio management and financial analytics.

Chapter 2: Company Background – This chapter explains Infosys Springboard and its important role in providing industry-oriented learning opportunities through virtual internships, certification programs, and technical skill development initiatives. It highlights the platform’s contribution toward supporting students and learners by offering practical exposure to modern technologies, real-world projects, and professional development activities. The chapter also describes the learning

environment, internship structure, training methodology, and educational resources provided through the Infosys Springboard platform. It explains how the internship program helped learners improve their technical knowledge, problem-solving abilities, and practical implementation skills through hands-on project development and industry-relevant assignments.

Chapter 3: Technology Stack – This chapter explains the various technologies, frameworks, and tools used during the development of the CRYPTOPORT application. It highlights the use of Python, Streamlit, data visualization libraries, financial analytics concepts, AI-based analysis modules, and cloud deployment technologies along with their applications in the project implementation process

Chapter 4: Technical Training and Assignment – This chapter summarizes the technical learning and practical assignments completed during the internship. It includes training related to Python programming, cryptocurrency market analysis, dashboard development, portfolio management, forecasting methods, risk analysis, data handling, and implementation of interactive financial analytics features within the application.

Chapter 5: Internship Project Contribution – This chapter explains the complete project development process and practical contributions made during the implementation of the crypto portfolio manager project. It includes the development of modules such as market dashboards, AI trading signals, portfolio analysis, forecasting systems, risk assessment tools.

Chapter 2

COMPANY BACKGROUND

Infosys Springboard is a digital learning and virtual internship platform launched by Infosys with the objective of empowering students and learners through free access to high-quality technical education, industry-relevant training, and practical project-based learning opportunities. The platform was designed to bridge the gap between academic education and industrial requirements by providing learners with modern technological skills and real-world implementation experience. Infosys Springboard offers a wide range of learning programs in emerging technologies such as Artificial Intelligence, Machine Learning, Cybersecurity, Cloud Computing, Python Programming, Data Science, Web Development, Internet of Things (IoT), and Data Analytics. Through its structured courses, certification programs, and internship opportunities, the platform helps students improve both technical knowledge and professional competencies required in today's competitive technology industry.

The Infosys Springboard Virtual Internship Program provides students with hands-on exposure to real-world projects under mentor guidance. The internship environment encourages innovation, problem-solving, teamwork, self-learning, and practical implementation of technical concepts. Students are assigned industry-oriented projects that simulate real business scenarios and help them gain valuable software development experience. As part of the internship program, the assigned project was “Crypto Portfolio Manager – Group 2”, which focused on developing an AI-powered cryptocurrency portfolio management platform named CRYPTOPORT. The project aimed to assist users in monitoring cryptocurrency investments, analyzing market trends, evaluating investment risks, generating AI-based trading signals, and visualizing financial analytics through an interactive web application.

The internship environment provided continuous learning opportunities through project implementation, technical guidance, practical assignments, and self-paced learning modules. Exposure to real-time application development and financial analytics enhanced practical understanding of software engineering concepts, user interface design, data visualization, and AI-assisted decision-making systems. Infosys Springboard also promotes continuous learning and skill enhancement by encouraging learners to explore advanced technologies and industry-standard development practices. The platform's focus on practical knowledge, innovation, and career

readiness makes it an important initiative for students preparing for professional roles in the software and technology industry. The internship experience at Infosys Springboard played a significant role in improving technical expertise, analytical thinking, project development capability, and professional confidence while providing valuable exposure to modern software development methodologies and fintech-based application implementation.

2.1 Services Provided

Infosys Springboard provides a wide range of digital learning services, virtual internship opportunities, and skill development programs aimed at improving technical knowledge and enhancing employability among students and learners. The platform focuses on delivering industry-oriented education through practical learning modules, certification courses, and real-world project exposure. One of the major services provided by Infosys Springboard is online technical training in emerging technologies. The platform offers learning resources and courses in domains such as Artificial Intelligence, Machine Learning, Cybersecurity, Cloud Computing, Python Programming, Data Science, Web Development, Internet of Things (IoT), and Data Analytics. These programs are designed to help learners build strong technical foundations and gain practical implementation skills.

Another important service provided by the platform is the Virtual Internship Program, where students are assigned real-world projects under the guidance of mentors and industry professionals. These internships help learners understand software development processes, project implementation strategies, teamwork, and professional work environments. The internship program encourages hands-on learning and practical application of technical concepts through industry-based assignments and project development activities. Infosys Springboard also provides certification programs that help students validate their technical knowledge and improve their career opportunities. The certifications are designed according to industry requirements and focus on both theoretical understanding and practical implementation skills.

In addition to technical education, the platform promotes self-paced learning and continuous skill development through interactive learning materials, quizzes, assignments, and project-based activities. Learners can access educational resources anytime and improve their knowledge according to their learning pace and interests.

The platform further supports innovation and career readiness by helping students develop problem-solving abilities, analytical thinking, communication skills, and professional ethics. Through its collaborative learning environment and industry-focused approach, Infosys Springboard prepares students for real-world software development and technology-related careers. During the internship, these services played an important role in the successful development of the “Crypto Portfolio Manager” project by providing technical guidance, learning resources, and practical exposure to modern software development methodologies.



Fig 2.1: Services Provided

Fig 2.1 illustrates the various services and opportunities provided by Infosys Springboard for learners, students, and professionals. The platform focuses on three major areas: Learn, Prepare, and Connect, which together help users build technical knowledge, improve career readiness, and explore professional opportunities. Under the learning section, Infosys Springboard offers online courses, expert-led training, certifications, learning paths, hands-on practice, and assessments that enable users to gain industry-relevant skills in multiple domains. These resources are designed to support self-paced learning and practical skill development through projects, labs, and interactive content.

In addition to technical learning, Infosys Springboard also focuses on overall career development by providing various professional growth opportunities such as career guidance, resume-building support, interview preparation, soft skills training, aptitude development, and portfolio-building activities. These learning resources help students improve both their technical knowledge and professional skills required in real-world corporate environments.

2.2 Vision, Mission & Certifications

Infosys Springboard is a digital learning platform that focuses on providing accessible and industry-oriented education to students and learners. The platform aims to help individuals improve their technical knowledge, practical skills, and professional capabilities through modern learning programs and hands-on training opportunities. Its vision is centered on supporting learners in becoming industry-ready professionals by offering quality educational resources, technical courses, and real-world project exposure. The mission of Infosys Springboard is to create an inclusive learning environment where students can continuously develop their knowledge and skills in emerging technologies.

The platform encourages practical learning, innovation, critical thinking, and problem-solving by providing access to virtual internships, certification programs, technical courses, and project-based learning activities. Through these initiatives, learners are able to strengthen both their academic understanding and practical implementation abilities. Infosys Springboard also places strong emphasis on experiential learning through real-time projects and technical assignments. Students are encouraged to apply theoretical concepts in practical development environments, which helps improve confidence, technical expertise, and understanding of professional software development workflows. The internship programs offered by the platform expose learners to modern technologies, industry practices, and real-world application development processes.

The platform provides certification programs in various technical domains such as Artificial Intelligence, Machine Learning, Cloud Computing, Cybersecurity, Python Programming, Data Science, and Web Development. These certification programs help learners validate their technical skills and improve their career opportunities in the technology sector. By combining learning resources, practical exposure, and industry-focused training, Infosys Springboard plays an important role in supporting technical education and professional growth for students and aspiring developers. The platform further supports career development by offering certifications in various domains such as Artificial Intelligence, Machine Learning, Python Programming, Data Science, Cloud Computing, Cybersecurity, and Web Development. These certification programs help learners validate their technical skills and improve their career opportunities in the IT industry.

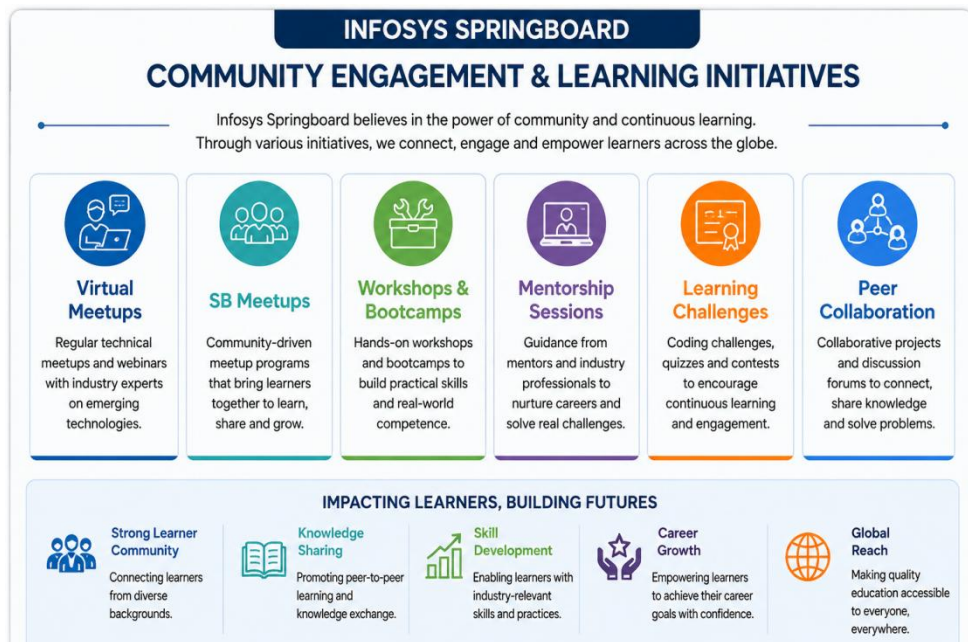


Fig 2.2: Community Engagement and Learning Initiatives

Fig 2.3 illustrates the community engagement and learning initiatives conducted through the Infosys Springboard platform. The figure highlights various activities such as virtual meetups, mentorship sessions, workshops, bootcamps, peer collaboration, and learning challenges organized to enhance students' technical and professional skills. These initiatives encourage knowledge sharing, practical learning, teamwork, and continuous skill development among learners from different educational backgrounds. The platform's collaborative learning environment helps students improve problem-solving abilities, gain industry exposure, and prepare for future career opportunities in the technology sector.

2.3 Community Engagement & SB Meetups

Infosys Infosys Springboard strongly believes in the importance of community engagement, collaborative learning, and continuous skill development. The platform encourages students and learners to actively participate in various technical initiatives, workshops, mentorship programs, and virtual learning communities that help improve both technical and professional competencies. Springboard strongly promotes community engagement and collaborative learning through its digital education initiatives, virtual training programs, and industry-oriented internship opportunities. The platform believes that continuous learning and knowledge sharing are essential for developing skilled professionals capable of adapting to rapidly evolving technologies and industry requirements.

The platform regularly conducts virtual workshops, webinars, technical sessions, coding activities, and mentorship programs to help learners stay updated with modern technologies and current industry trends. These initiatives create opportunities for students to gain insights from industry experts, improve problem-solving abilities, and understand real-world software development practices. Another important aspect of Infosys Springboard is its focus on collaborative project development and innovation. The platform also encourages self-paced learning and continuous skill enhancement by providing flexible learning opportunities that allow students to improve their knowledge. Through these community engagement and learning initiatives, Infosys Springboard helps learners build confidence, technical expertise, and career readiness for future opportunities in the technology industry.

During the internship, these initiatives played an important role in improving practical understanding of software development and analytical application development workflows. The interactive learning environment also encouraged independent learning, creativity, and practical implementation of technical concepts through project-based activities. Participating in these programs helped improve communication skills, teamwork, and confidence in handling real-world technical challenges. Overall, the learning initiatives provided by Infosys Springboard significantly contributed toward both technical growth and professional development during the internship experience.

Chapter 3

TECHNOLOGY STACK

The technology stack plays a very important role in the successful development of any modern software application. It includes the programming languages, frameworks, libraries, tools, and platforms used during the design, development, testing, and deployment phases of a project. Choosing the right technologies helps improve application performance, scalability, security, maintainability, and overall user experience. For the development of the AI Powered Crypto Portfolio Manager project, several modern Python-based technologies and analytical tools were used to build an efficient, interactive, and data-driven cryptocurrency management platform. The project required functionalities such as real-time cryptocurrency market analysis, portfolio management, forecasting, portfolio risk evaluation, AI-based trading insights, and interactive dashboard visualization. To implement these features effectively, different technologies and frameworks were integrated throughout the development process.

Python was used as the core programming language because of its simplicity, flexibility, and strong support for data analysis and financial computation. The Streamlit framework was used for developing interactive dashboards and creating a responsive web-based user interface. Libraries such as Pandas and NumPy were used for cryptocurrency data processing, financial analysis, numerical computations, and portfolio calculations. Plotly was integrated to develop interactive charts, forecasting graphs, portfolio allocation visuals, and analytical dashboards for better financial understanding and market visualization. Various analytical and AI-based modules were also implemented within the application to support cryptocurrency forecasting, portfolio analysis, trading signal generation, and portfolio risk assessment functionalities.

These technologies helped generate predictive insights and improved the analytical capabilities of the platform. Interactive charts and graphical representations further enhanced the user experience by simplifying complex financial information and presenting market trends in a visually understandable format. The application was deployed using cloud-based hosting services such as Streamlit Community Cloud, allowing users to access the platform directly through a web browser without requiring complex local installation. Modern development practices, modular coding techniques,

debugging methods, and version control management using Git and GitHub were also followed throughout the project lifecycle to ensure better scalability, maintainability, and organized project development. This chapter provides a detailed explanation of the major technologies, frameworks, libraries, and tools used during the internship project along with their important features and practical applications within the CRYPTOPORT system.

3.1 Python

Python is a high-level, interpreted, and versatile programming language widely used in software development, data analysis, artificial intelligence, machine learning, automation, and web application development. Due to its simple syntax, extensive library support, and strong community ecosystem, Python has become one of the most popular programming languages in the technology industry. In the “Crypto Portfolio Manager” project, Python served as the core programming language for implementing various functionalities related to cryptocurrency market analysis, portfolio management, forecasting, risk assessment, and data visualization.

The language provided an efficient and flexible environment for handling financial datasets, performing analytical operations, and developing AI-assisted features within the application. Python offers excellent support for financial analytics and data science applications through libraries such as Pandas, NumPy, Plotly, and other analytical tools. These libraries helped in processing cryptocurrency market data, calculating portfolio performance, generating analytical insights, and visualizing financial trends through interactive charts and dashboards. One of the major advantages of Python is its readability and ease of implementation, which simplifies complex programming tasks and improves development efficiency.

The language also supports modular programming and easy integration with modern web frameworks, making it highly suitable for developing scalable, maintainable, and efficient applications. During the internship project, Python was extensively used for implementing cryptocurrency market tracking systems, AI-based trading signals, forecasting modules, portfolio risk analysis calculations, and interactive dashboard functionalities within the CRYPTOPORT platform. Python enabled smooth integration between analytical models, financial computation modules, and the Streamlit web application interface, resulting in an efficient and user-friendly cryptocurrency portfolio management system.



Fig 3.1: Python Programming Language

Fig 3.1 shows the Python programming language logo, which represents the primary development technology used throughout the implementation of the CRYPTOPORT project. Its flexibility and extensive library support helped simplify the development process and improved the overall analytical capabilities of the application. Python made it easier to manage real-time cryptocurrency data processing, visualization, and forecasting operations within a single integrated environment. The language also improved development efficiency by supporting reusable code structures, modular implementation, and faster analytical computation, which significantly contributed to the successful development of the CRYPTOPORT application.

3.1.1 Key Features of Python

Python provides several powerful features that make it one of the most preferred programming languages for software development, artificial intelligence, financial analytics, and web application development. Its simplicity, flexibility, and extensive library support played an important role in the successful development of the Crypto Portfolio Manager project. **Simple and Readable Syntax :** Python has a clean and easy-to-understand syntax, which allows developers to write programs efficiently with fewer lines of code. Its readability improves code maintenance, debugging, and overall development productivity. **Extensive Library Support:** Python provides a large collection of built-in and external libraries for data analysis, visualization, machine learning, and web development.

Libraries such as Pandas, NumPy, Plotly, and Matplotlib were useful for handling cryptocurrency data, performing calculations, and generating interactive charts within the project. **Platform Independence:** Python is a platform-independent programming language, meaning applications developed in Python can run on different operating systems such as Windows, Linux, and macOS without major modifications. This feature improves application portability and flexibility.

Integration with Web Frameworks: Python supports various web frameworks and dashboard development tools such as Streamlit, Flask, and Django. In this project, Python was integrated with the Streamlit framework to develop an interactive cryptocurrency portfolio management dashboard. **Fast Development and Prototyping:** Python allows rapid application development of new features. This helped in developing modules such as portfolio tracking, forecasting dashboards, AI chatbot integration, and analysis systems during the internship. **Strong Community Support:** Python has a large global developer community that continuously contributes libraries, documentation, and technical resources. This support helps developers solve technical issues efficiently and improve application development processes.

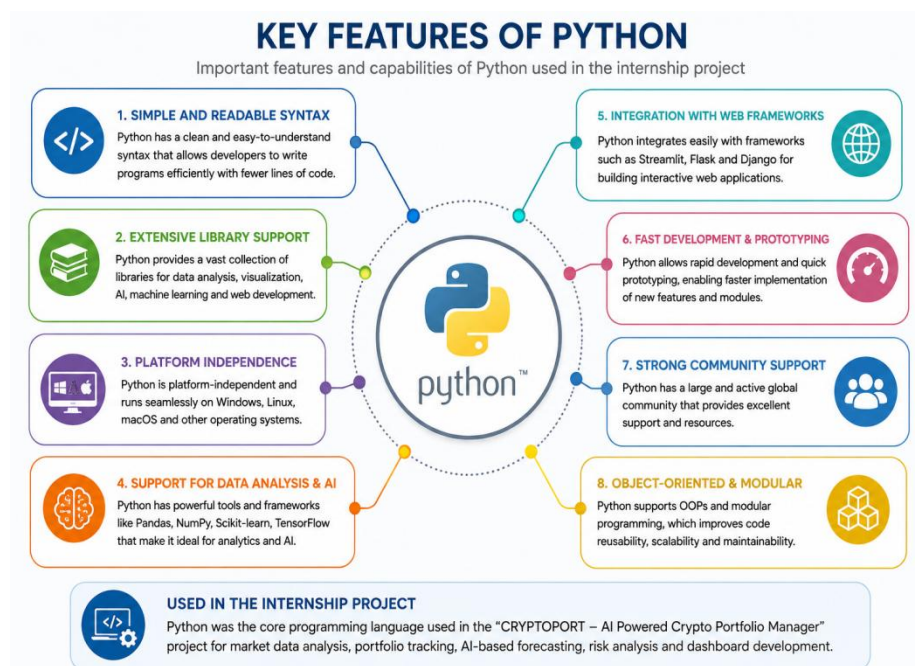


Fig 3.2: Key Features of Python

Fig 3.2 shows the important features and capabilities of the Python programming language used during the internship project. **Support for Data Analysis and AI:** Python is widely used in Artificial Intelligence, Machine Learning, and Data Analytics because of its powerful analytical libraries and frameworks. This feature made Python highly suitable for implementing AI-based forecasting, trading signal generation, and risk analysis functionalities in the CRYPTOPORT system. **Object-Oriented and Modular Programming:** Python supports object-oriented programming concepts and modular development practices, which improve code reusability, scalability, and maintainability. These features helped organize the project modules efficiently during the development of the CRYPTOPORT application.

3.1.2 Use Case in My Internship

During my internship, Python served as the core technology for developing the CRYPTOPORT – AI Powered Crypto Portfolio Manager application. The project required features such as real-time cryptocurrency market analysis, portfolio tracking, forecasting, portfolio risk evaluation, and interactive dashboard visualization. Python provided an efficient and flexible environment for implementing these functionalities and integrating different analytical modules within a single platform. One of the major applications of Python during the project was cryptocurrency market data analysis. I used Python libraries such as Pandas and NumPy to process cryptocurrency datasets, calculate asset performance, analyze historical market trends, and perform portfolio-related financial calculations efficiently. These libraries helped manage large amounts of financial data and generated meaningful analytical insights for users.

Python was also used to implement AI-based trading signal modules within the application. Using analytical logic and cryptocurrency market trend analysis, the system generated Buy, Hold, and Sell recommendations based on market conditions, forecasting results, and portfolio performance. These intelligent trading insights helped users better understand investment opportunities and market behavior. Another important implementation area was portfolio risk analysis and financial evaluation. I developed functionalities that calculated portfolio volatility, profit and loss values, asset allocation percentages, and overall portfolio risk scores. These analytical features helped users evaluate investment stability, monitor portfolio health, and understand financial exposure more effectively.

Using the Streamlit framework integrated with Python, I created responsive and interactive dashboards for displaying live cryptocurrency prices, market trends, forecasting results, portfolio analytics, and risk analysis visualizations in real time. Interactive user interface components and dynamic dashboards improved accessibility and enhanced overall user experience within the platform. For data visualization, I implemented graphical representations such as line charts, forecasting graphs, portfolio allocation visuals, and analytical dashboards using Plotly and other visualization libraries. These visualizations simplified complex financial information and helped users interpret cryptocurrency market behavior more effectively. Python was further used for implementing forecasting and predictive analysis modules based on historical cryptocurrency trends and calculations. These systems helped predict market movements and supported users in making data-driven investment decisions.

I integrated an AI- powered chatbot assistant within the application to provide cryptocurrency-related guidance, portfolio insights, and basic investment support through an interactive chat interface. Finally, Python and Streamlit were used for deploying the CRYPTOPORT application on Streamlit Community Cloud, enabling users to access the platform through a web browser with a smooth, responsive, and professional interface. Overall, working with Python during the internship significantly improved my practical understanding of financial analytics, AI integration, dashboard development, cloud deployment, and real-world fintech application development.

3.1 Streamlit Framework

Streamlit is an open-source Python framework used for developing interactive web applications and data-driven dashboards with minimal frontend coding. It is widely used for machine learning applications, financial analytics dashboards, data visualization systems, and AI-based web applications because of its simplicity, flexibility, and rapid development capabilities. In the “Crypto Portfolio Manager” project, Streamlit served as the primary framework for developing the interactive user interface and dashboard system. The framework enabled smooth integration between Python-based analytical modules and the web application frontend, allowing real-time visualization of cryptocurrency market data, portfolio analytics, forecasting results, and AI-generated trading signals.

One of the major advantages of Streamlit is its ability to convert Python scripts directly into interactive web applications without requiring extensive knowledge of frontend technologies such as HTML, CSS, or JavaScript. This significantly simplified the development process and allowed faster implementation of project modules during the internship period. Streamlit provides several built-in components such as interactive charts, buttons, forms, sliders, dropdown menus, tables, and sidebar navigation systems. These features were extensively used in the CRYPTOPORT project to create a professional and user-friendly interface for cryptocurrency portfolio monitoring and financial analysis.

The below Fig 3.2 shows the Streamlit framework logo and its role in the development of the Crypto Portfolio Manager project. Streamlit is an open-source Python framework used for building interactive web applications and analytical dashboards with minimal frontend coding.



Fig 3.3: Streamlit Framework

The framework also supports integration with data visualization libraries such as Plotly, Matplotlib, and Pandas, making it highly suitable for financial dashboard development. Using Streamlit, various analytical sections such as Market Dashboard, AI Signals, Forecasting System, Risk Analysis, Portfolio Management, and AI Chat Assistant were implemented successfully within the project. Another important feature of Streamlit is its support for rapid deployment and cloud hosting. The CRYPTOPORT application was deployed using Streamlit Community Cloud, allowing users to access the platform online through a web browser without requiring local installation or complex configurations. During the internship, Streamlit played a vital role in improving development efficiency, simplifying dashboard implementation, and providing an interactive user experience for cryptocurrency portfolio analysis and AI-based investment management.

3.2.1 Key Features of Streamlit

Streamlit provides several powerful and user-friendly features that make it highly suitable for developing interactive dashboards and data-driven web applications. During the development of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project, these features played a major role in building a responsive, efficient, and visually interactive cryptocurrency analytics platform. One of the major advantages of Streamlit is its simple and fast development process. The framework allows developers to build complete web applications directly using Python without requiring additional frontend technologies such as HTML, CSS, or JavaScript. This significantly reduced development complexity and improved implementation speed during the project.

Streamlit also provides various built-in interactive user interface components such as buttons, forms, sliders, tables, dropdown menus, and sidebars. These components were used to create dynamic dashboards and improve user interaction within the CRYPTOPORT application. The dashboard allowed users to navigate

between modules such as Market Dashboard, Portfolio Management, Forecasting System, Risk Analysis, and AI Assistant easily. Another important feature of Streamlit is its support for real-time updates. The framework automatically refreshes the application whenever data or code changes occur, making it highly suitable for live cryptocurrency market analysis and portfolio monitoring systems. This helped the application continuously display updated cryptocurrency prices, forecasting insights, and portfolio analytics. Streamlit also integrates smoothly with popular Python libraries such as Pandas, NumPy, Plotly, and Matplotlib. These integrations helped implement financial data analysis, portfolio calculations, interactive charts, forecasting systems, and graphical visualizations efficiently within the application dashboard.

The framework additionally provides strong support for interactive data visualization. Using Streamlit with Plotly, the project displayed cryptocurrency price trends, candlestick charts, profit/loss graphs, asset allocation charts, and forecasting analytics in an interactive graphical format that improved user understanding and financial analysis. Another advantage of Streamlit is that it is lightweight, open-source, and easy to deploy. The application was deployed using Streamlit Community Cloud, which enabled online access through web browsers without requiring complex server configuration. Streamlit greatly simplified the development of professional and responsive dashboards while improving the efficiency, usability, and analytical capabilities of the CRYPTOPORT platform.

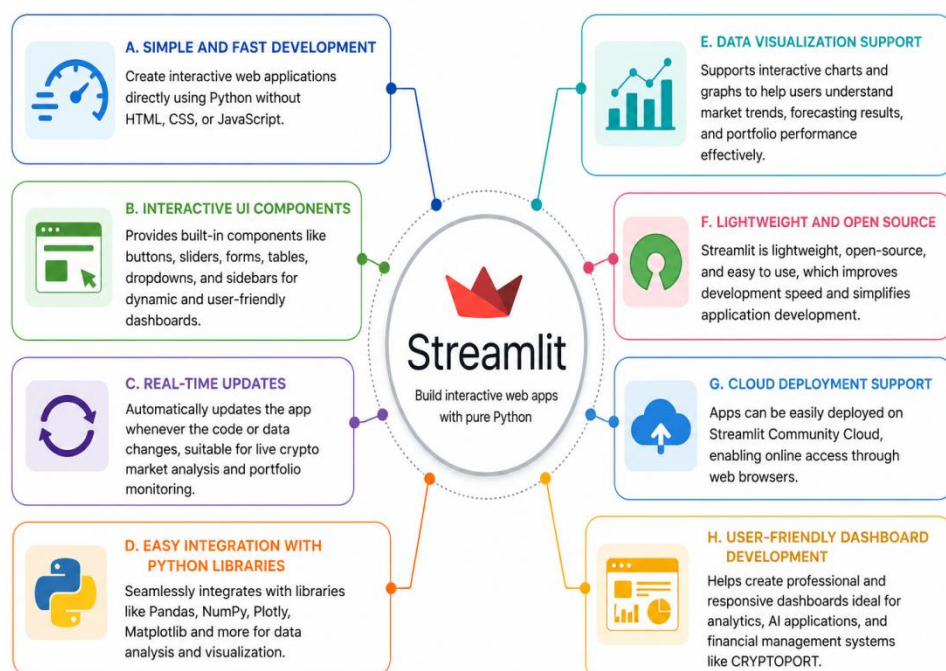


Fig 3.4: Key Features of Streamlit Framework

Fig 3.5 illustrates the important features of the Streamlit framework used during the development of the CRYPTOPORT project. The figure highlights features such as fast development, interactive UI components, real-time updates, data visualization support, cloud deployment, and seamless integration with Python libraries for building professional analytical dashboards.

3.2.2 Application Workflow in My Internship

During my internship, Streamlit was used as the main framework for developing the interactive dashboard and frontend system of the CRYPTOPORT – AI Powered Crypto Portfolio Manager application. The framework helped connect Python-based analytical modules directly with the user interface, making the development process faster, simpler, and more efficient. Using Streamlit, I was able to build a professional cryptocurrency management platform with real-time analytics, interactive dashboards, and user-friendly navigation features. One of the major contributions during development was the implementation of forecasting and trading signal modules.

Streamlit was used to display AI-generated Buy, Hold, and Sell recommendations along with forecasting charts, predictive analytics, and market trend visualizations. These features helped users understand cryptocurrency market behavior and make better investment decisions based on analytical insights. To improve user experience and dashboard interaction, several Streamlit components such as buttons, dropdown menus, sidebars, forms, tables, and interactive charts were implemented throughout the application. These interface components helped create smooth navigation between different modules such as the Dashboard, Risk Analysis, Forecasting Dashboard, and AI Assistant. Streamlit visualization features were also used to develop the Portfolio Risk Analysis dashboard.

The dashboard displayed volatility scores, portfolio risk levels, asset performance analytics, and investment stability indicators through organized graphical layouts and interactive financial charts. This helped users better understand portfolio risk and market exposure. Another important implementation was the integration of the AI Chat Assistant within the Streamlit environment. The chatbot interface allowed users to interact with the system, ask cryptocurrency-related questions, receive portfolio insights, and access investment guidance through a simple conversational interface. Overall, Streamlit played a major role in simplifying dashboard development, improving real-time visualization, and enhancing the usability and analytical capabilities of the CRYPTOPORT platform.

3.3 Cryptocurrency Data Analysis & Visualization Libraries

Cryptocurrency data analysis and visualization libraries played a major role in the development of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. These libraries helped in processing financial datasets, analyzing cryptocurrency market trends, generating portfolio insights, and creating interactive graphical dashboards for users. During the project development, Python libraries such as Pandas, NumPy, and Plotly were widely used to handle cryptocurrency-related data efficiently and improve the analytical capabilities of the application. Pandas was mainly used for organizing and processing cryptocurrency market datasets. It helped manage tabular financial information, clean market data, calculate portfolio performance, and perform different analytical operations required for real-time cryptocurrency analysis. Using Pandas made it easier to work with large datasets and generate meaningful portfolio insights within the dashboard.

NumPy was used for numerical and mathematical computations throughout the project. It supported calculations related to forecasting, volatility analysis, profit/loss evaluation, portfolio returns, and risk assessment. The library improved the speed and accuracy of financial calculations and helped process cryptocurrency market information efficiently. Plotly was used to create interactive charts and graphical visualizations for the dashboard. Various visual analytics such as cryptocurrency trend graphs, candlestick charts, forecasting plots, portfolio allocation charts, and profit/loss analysis graphs were developed using Plotly. These visualizations made the dashboard more interactive, visually appealing, and easier for users to understand.

The integration of analytical libraries such as Pandas, NumPy, and Plotly with the Streamlit framework significantly improved the responsiveness and overall user experience of the CRYPTOPORT application. Real-time cryptocurrency market data could be processed, analyzed, and displayed dynamically through interactive dashboards and financial visualizations. Various visual analytics such as cryptocurrency trend graphs, candlestick charts, forecasting plots, portfolio allocation charts, and profit/loss analysis graphs were developed using Plotly to help users better understand market behavior and portfolio performance. These analytical and visualization libraries contributed greatly to the successful implementation of market dashboards, portfolio management systems, AI-based trading signals, forecasting modules, and risk analysis functionalities within the CRYPTOPORT platform.

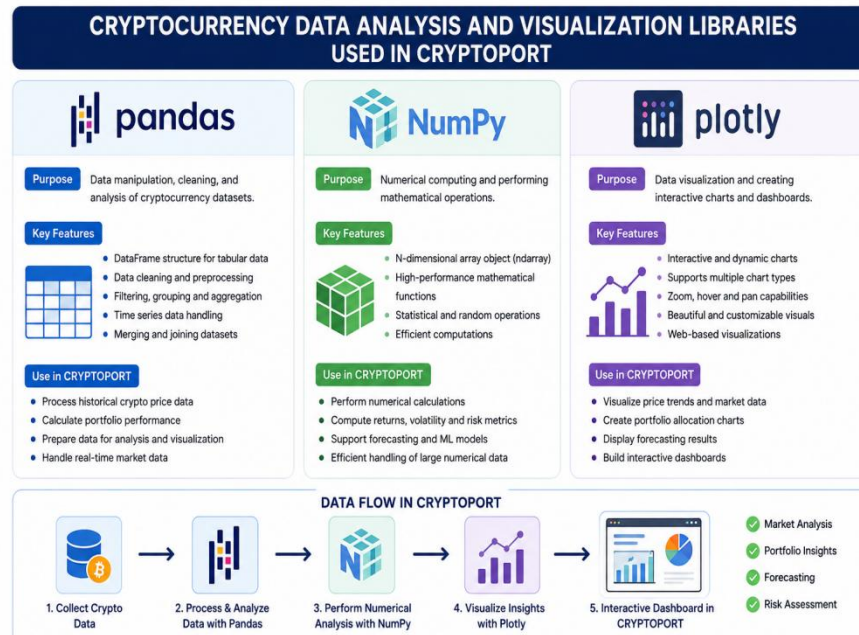


Fig 3.5: Data Analysis and Visualization Libraries

Fig 3.6 illustrates the major cryptocurrency data analysis and visualization libraries used in the development of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. The figure highlights the role of important Python libraries such as Pandas, NumPy, and Plotly in handling financial data processing, numerical analysis, and interactive dashboard visualization within the application. Pandas was mainly used for organizing, cleaning, and processing cryptocurrency market datasets. It helped manage tabular financial data, perform portfolio calculations, analyze market trends, and prepare data for forecasting and visualization operations. The library made it easier to work with real-time cryptocurrency information efficiently within the dashboard system.

3.3.1 Key Features of Pandas, NumPy, and Plotly

The CRYPTOPORT – AI Powered Crypto Portfolio Manager project made extensive use of Python libraries such as Pandas, NumPy, and Plotly to support cryptocurrency data management, financial analysis, forecasting, and dashboard visualization functionalities. These libraries formed the core analytical foundation of the application and played an important role in processing real-time cryptocurrency market information efficiently. Pandas was mainly used for handling and analyzing cryptocurrency datasets within the project. The library provided powerful DataFrame structures that made it easier to organize, process, and manage large amounts of financial data.

Pandas also supported time-series analysis, which was very useful for monitoring cryptocurrency price movements, tracking market trends, and analyzing historical market performance over time. Its ability to efficiently manage portfolio records made it one of the most important components of the application. NumPy was used for performing numerical and mathematical computations required for cryptocurrency analysis and forecasting operations. The library helped process large datasets quickly and efficiently through its high-performance array and computational capabilities.

NumPy was mainly used for statistical calculations, portfolio return analysis, volatility calculations, forecasting models, and financial risk assessment techniques. It also improved the speed and performance of analytical processing throughout the application, making financial calculations more accurate and efficient. Plotly was used to create interactive charts and professional financial visualizations within the dashboard system. The library supported various graphical representations such as line charts, bar graphs, pie charts, candlestick charts, forecasting plots, and portfolio allocation visuals.

Plotly also provided advanced interactive features such as zooming, hovering, panning, and dynamic chart updates, which improved user engagement and data interpretation within the application dashboard. Together, Pandas, NumPy, and Plotly significantly improved the analytical capabilities, responsiveness, and usability of the CRYPTOPORT platform. Their combined functionality enabled efficient cryptocurrency data processing, portfolio analysis, forecasting, risk evaluation, and interactive dashboard visualization, contributing greatly to the development of a complete and user-friendly cryptocurrency investment management system.

3.3.2 Application Workflow in My Internship

During my internship, I worked extensively with the Pandas, NumPy, and Plotly libraries while developing the Crypto Portfolio Manager (CRYPTOPORT) project. These libraries formed the core of the application's data processing, financial analysis, and visualization capabilities. They enabled the system to efficiently manage cryptocurrency market information, perform complex calculations, and present insights through an interactive and user-friendly dashboard. A major part of the application involved processing cryptocurrency market data. Pandas was used to collect, clean, organize, and analyze large datasets obtained from various cryptocurrency sources. By handling data preprocessing and transformation tasks, the library ensured that market information remained accurate and structured for further analysis.

Its support for time-series data was particularly useful for monitoring price fluctuations and identifying market trends over different periods. The application also focused on portfolio management by maintaining detailed records of users' cryptocurrency investments. Information such as asset names, quantities held, purchase prices, current market values, transaction history, and profit or loss calculations was efficiently managed using Pandas DataFrames. This structured approach allowed users to track their investments and evaluate portfolio performance in real time. To support investment decision-making, the system stored and analyzed forecasting results, market trend indicators, risk assessment scores, AI-generated trading signals, and portfolio performance metrics.

NumPy was extensively used for performing statistical calculations, numerical analysis, and financial computations. Its high-performance array-processing capabilities helped improve the speed and accuracy of forecasting models and risk evaluation processes. Another important aspect of the project was performing various data operations within the application. Pandas and NumPy worked together to calculate portfolio values, determine profit and loss figures, update market prices, generate forecasting outputs, and produce risk analysis reports. These operations were integrated into the Streamlit dashboard, enabling users to access analytical results instantly and interact with the data more effectively. The application also provided dynamic querying and filtering features to enhance data accessibility.

Users could retrieve portfolio information for specific cryptocurrencies, analyze market trends based on selected time periods, view forecasting results and AI trading signals, and monitor portfolio performance through customized filters. Pandas offered powerful filtering, grouping, and aggregation functionalities that made these features efficient and responsive. To improve data interpretation and user engagement, Plotly was used to create interactive visualizations throughout the dashboard. Various charts, including portfolio allocation graphs, market trend visualizations, performance comparisons, and forecasting charts, were developed using Plotly.

Interactive features such as zooming, hovering, and real-time chart updates allowed users to explore financial data more effectively and gain deeper insights into market behavior. Information such as asset names, quantities held, purchase prices, current market values, transaction history, and profit or loss calculations was efficiently managed using Pandas DataFrames. The integration of Pandas, NumPy, and Plotly played a crucial role in the development of the CRYPTOPORT application.

3.4 AI-Based Market Forecasting and Risk Analysis

AI-Based Market Forecasting and Risk Analysis played an important role in the development of the “Crypto Portfolio Manager” project. These modules were designed to help users analyze cryptocurrency market trends, predict possible future price movements, evaluate investment risks, and make better financial decisions using data-driven insights. The forecasting system was developed using Python analytical techniques and historical cryptocurrency market data. The system generated predictive insights regarding possible future market conditions. These forecasting features helped users understand market direction and investment opportunities more effectively.

The AI-based trading signal module was implemented to provide Buy, Hold, and Sell recommendations based on market trend analysis and portfolio conditions. This functionality assisted users in identifying potential investment actions using analytical calculations and predictive logic integrated within the application. Risk analysis was another important component of the project. The system evaluated cryptocurrency portfolio risk by analyzing factors such as asset volatility, portfolio diversification, market fluctuations, and investment performance. Risk scores and portfolio health indicators were generated to help users understand the stability and exposure level of their investments.

Interactive visualizations and forecasting charts were implemented using graphical libraries to display trend analysis, market behavior, and risk metrics in a clear and understandable manner. These visual analytical tools improved user understanding of financial data and enhanced the overall decision-making experience within the platform. The forecasting were fully integrated into the Streamlit dashboard, enabling users to access analytical insights, market predictions, and risk evaluation reports in real time through a professional and interactive web interface. These forecasting features helped users understand market direction and investment opportunities more effectively.

During the internship, these AI-based analytical systems significantly improved the functionality and intelligence of the CRYPTOPORT application by combining financial analysis, predictive modeling, and interactive visualization techniques. Users could retrieve portfolio information for specific cryptocurrencies, analyze market trends based on selected time periods, view forecasting results and AI trading signals, and monitor portfolio performance through customized filters.

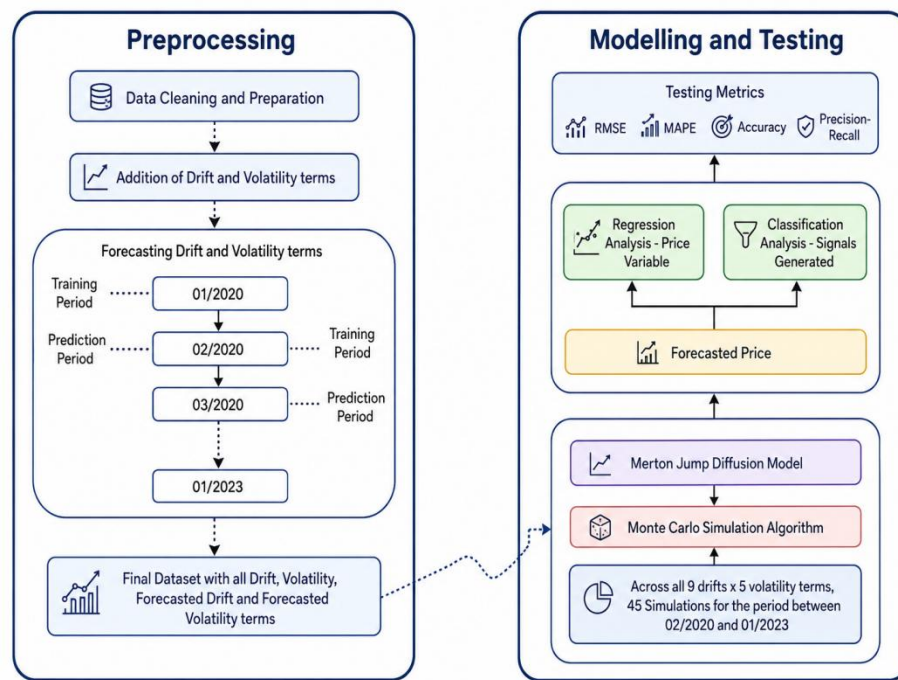


Fig 3.6: AI-Based Market Forecasting

Fig 3.4 illustrates the AI-Based Market Forecasting and Risk Analysis System developed in the CRYPTOPORT application. The system analyzes real-time and historical cryptocurrency market data to identify market trends, predict future price movements, and evaluate investment risks. It also provides forecasting insights and risk analysis through interactive charts and visual dashboards, helping users better understand market behavior and make more informed cryptocurrency investment decisions.

3.4.1 Role of AI and Analytical Models in Development

Artificial Intelligence and analytical models played a major role in the development of the AI Powered Crypto Portfolio Manager project. These technologies were used to analyze cryptocurrency market behavior, generate forecasting insights, evaluate investment risks, and support users in making better financial decisions based on market trends and portfolio performance. The AI-based analytical system worked by processing historical cryptocurrency market data, price fluctuations, and trend patterns to identify possible future market movements.

By analyzing large amounts of financial data, the system was able to provide intelligent trading recommendations for different cryptocurrencies. Analytical models were also implemented to calculate important financial metrics such as portfolio performance, asset volatility, profit/loss values, investment diversification, and

portfolio risk levels. These calculations helped users better understand the stability and overall performance of their cryptocurrency investments. One of the most important AI functionalities in the project was the generation of Buy, Hold, and Sell signals based on forecasting analysis and market trend evaluation. These intelligent recommendations helped users identify potential investment opportunities and manage portfolio risks more effectively.

The forecasting system additionally provided graphical trend analysis and predictive visualization dashboards that allowed users to understand future market behavior more easily. Interactive charts and financial visualizations simplified complex cryptocurrency analysis and improved overall user understanding. AI and analytical models also improved automation within the platform by reducing manual calculations and continuously generating real-time insights. Their integration enhanced the intelligence, and analytical capabilities of the application, making it a more interactive and data-driven cryptocurrency investment management system.

3.4.2 Practical Implementation in My Internship

During my internship, I worked on implementing AI-driven forecasting and analytical features within the Crypto Portfolio Manager (CRYPTOPORT) application. These features were designed to help users understand cryptocurrency market behavior, evaluate investment risks, and make more informed financial decisions. By combining historical market data with analytical models, the application provided intelligent insights into market trends and portfolio performance. A key aspect of the system was market trend analysis, where historical cryptocurrency prices and market movements were analyzed to identify patterns and generate meaningful financial insights. The application continuously monitored price fluctuations and market performance, enabling users to better understand market conditions and potential future trends.

The project also incorporated an AI-based trading signal mechanism that generated Buy, Hold, and Sell recommendations. These recommendations were based on market analysis, forecasting results, and portfolio conditions. The objective was to provide users with actionable insights that could support their investment decisions and improve portfolio management strategies. To enhance predictive capabilities, AI-driven forecasting models were developed to estimate potential future cryptocurrency price movements. These models utilized historical market data to identify trends and generate forecasts. The forecasting results helped users evaluate possible market

scenarios and plan their investment strategies accordingly. Risk analysis was another important component of the application. The system calculated various risk indicators, including asset volatility, portfolio diversification levels, and overall risk scores. These metrics allowed users to assess the stability of their investments and better understand the potential risks associated with different cryptocurrency assets.

To improve the presentation of analytical results, forecasting outputs, trend analyses, and portfolio performance metrics were displayed through interactive visualizations. These graphical representations made complex financial information easier to understand and allowed users to explore data more effectively. The AI-based analytical framework acted as a decision support system by helping users analyze market conditions, identify potential investment opportunities, and manage portfolio risks. All forecasting, trading signal generation, and risk analysis functionalities were integrated into the Streamlit dashboard, providing a centralized and user-friendly platform for cryptocurrency analytics and investment management.

3.5 Git and GitHub (Version Control System)

Git and GitHub are important tools used in modern software development for version control, project management, and collaborative coding. These tools help developers manage code changes, track project progress, maintain multiple versions of applications, and work efficiently in development environments. During the development of the “Crypto Portfolio Manager” project, Git and GitHub played a vital role in maintaining an organized codebase and managing the overall development workflow. These tools helped in tracking modifications, managing project files, maintaining backup versions of the application, and improving development efficiency throughout the internship.

Git is a distributed version control system that allows developers to record changes in source code and maintain project history. It enables safe development practices by allowing code modifications without affecting the stable version of the project. Git also provides branching and merging functionalities that simplify feature development and debugging processes. GitHub is a cloud-based platform built on Git that provides remote repository hosting, collaboration tools, issue tracking, code sharing, and project management capabilities. It allows developers to store project repositories online and collaborate efficiently with team members and mentors. Using Git and GitHub during the internship improved project organization, source code

management, debugging efficiency, and development workflow management. Regular commits, repository updates, and version tracking ensured that all project modules were maintained properly throughout the development lifecycle. These tools also helped in maintaining backup copies of the project and supported smooth deployment and project sharing activities. The integration of GitHub with development workflows significantly improved project monitoring and software maintenance during the implementation of the CRYPTOPORT platform.



Fig 3.7: Git and Github

Fig 3.1 illustrates Git and GitHub, which were important tools used during the internship for managing the development workflow of the CRYPTOPORT project. Git helped in tracking code changes, maintaining different versions of the project, and organizing the development process efficiently. GitHub provided a platform for storing the project repository, managing updates, and supporting collaboration during development. Together these tools helped maintain a structured codebase, improve software quality, simplify debugging and version management, and ensure smooth project development through controlled collaboration and regular code updates.

3.6 Key Features of Git and GitHub.

During the development of the Crypto Portfolio Manager (CRYPTOPORT) project, Git and GitHub played a vital role in managing source code, tracking project progress, and supporting collaborative development. These tools helped maintain an organized workflow, ensured code reliability, and facilitated efficient project management throughout the development lifecycle. Git was used as the primary version control system, allowing developers to maintain a complete copy of the project on their local machines. This distributed approach enabled offline development, faster code management, and enhanced protection against data loss. It also provided flexibility for developers to work independently while keeping the project synchronized.

One of the key advantages of Git was its ability to track changes through commits. Each commit acted as a snapshot of the project at a specific stage of development, making it easier to monitor modifications, review progress, identify issues, and restore previous versions when necessary. This feature was particularly useful for maintaining project stability and managing updates efficiently. Git's branching and merging capabilities allowed different features, improvements, and bug fixes to be developed separately without affecting the main codebase.

Once the changes were tested and verified, they could be merged into the primary branch, ensuring a smooth and structured development process. GitHub served as the central platform for hosting and managing the project repository. It enabled seamless collaboration by allowing developers to share code, synchronize updates, and work together on different aspects of the project. The platform also provided secure cloud-based storage, ensuring that project files were safely backed up and accessible whenever required. To maintain code quality, GitHub's pull request and code review features were utilized. These tools allowed code changes to be reviewed before being merged into the main repository, helping identify potential issues, improve code standards, and ensure overall software reliability.

This collaborative review process contributed to the development of a more stable and maintainable application. In addition, GitHub supported continuous development practices by enabling integration with deployment and automation workflows. This helped streamline updates, simplify maintenance tasks, and improve overall development efficiency throughout the project. Overall Git and GitHub significantly enhanced source code management, team collaboration, and application maintenance during the implementation of the CRYPTOPORT platform.

3.6.1 Application Workflow.

During my internship, Git and GitHub were used extensively throughout the development of the Crypto Portfolio Manager project. These tools played a crucial role in managing source code, tracking project progress, maintaining version history, and ensuring an organized development workflow. Their use helped improve project reliability, code management, and overall development efficiency. Git was primarily used for version control, allowing me to track changes made to the application's source code throughout the development process. Every modification, enhancement, and bug fix was recorded systematically, making it easier to monitor project progress and maintain a complete history of the application.

The project's source code was hosted on GitHub, which served as a centralized repository for storing and managing project files. GitHub provided secure cloud-based storage, ensuring that the codebase remained accessible, backed up, and protected from data loss. It also made repository management more convenient by keeping all project resources organized in a single location. Regular commits were performed to save development progress and document updates made to the application. Each commit acted as a checkpoint, allowing changes to be tracked accurately and making it easier to identify when specific features or modifications were introduced. This practice helped maintain a structured and traceable development process throughout the project. Git also proved valuable during debugging and troubleshooting activities.

To support efficient feature development, Git branches were used for implementing new functionalities independently from the main codebase. Different modules, including cryptocurrency forecasting, dashboard enhancements, AI-based trading signal generation, and portfolio analysis features, were developed and tested in separate branches before being merged into the primary project repository. This approach reduced the risk of introducing errors into the main application and enabled smoother development workflows. Git also proved valuable during debugging and troubleshooting activities. By maintaining a detailed history of code changes, it became easier to identify the source of issues and restore previous stable versions of the application whenever required. This significantly improved the efficiency of bug fixing and application maintenance.

3.6.2 Learning Outcomes and Benefits .

The use of Git and GitHub during my internship greatly improved my understanding of version control systems, collaborative software development, and professional project management practices. Working with these tools provided valuable hands-on experience and gave me practical exposure to how modern software projects are managed in real-world development environments. One of the major learning outcomes was gaining practical knowledge of source code management using Git. I learned how to create and manage repositories, track code changes, maintain commit histories, and restore previous versions of the application whenever needed.

This experience helped me understand the importance of maintaining an organized, secure, and reliable development workflow throughout the project lifecycle. The internship also improved my understanding of branching and merging

techniques. I gained experience in developing and testing new features in separate branches before integrating them into the main project. This approach helped reduce coding errors, improved project stability, and made the development process more efficient and manageable. By working with GitHub, I developed a better understanding of cloud-based repository hosting and collaborative development practices. I learned how to manage repositories online, synchronize project updates, organize source code properly, and maintain secure backups of project files. These practices highlighted the importance of centralized repository management and proper version control in professional software development environments.

Working with Git and GitHub enhanced my coding discipline, project organization skills, version control knowledge, and understanding of collaborative software development practices. The experience also improved my confidence in handling real-world development workflows and contributed significantly to the successful completion of the CRYPTOPORT project. Another important learning outcome during the internship was the improvement of my debugging and troubleshooting skills. Git's version tracking and commit history made it easier to identify coding issues, review previous implementations, and resolve errors efficiently during development.

This helped maintain the stability, performance, and reliability of the CRYPTOPORT application throughout the project lifecycle. Working with GitHub also improved my understanding of project documentation, repository organization, and deployment management. Maintaining a well-structured repository supported smooth development, simplified project maintenance, and made future updates and enhancements easier to manage. Overall, using Git and GitHub strengthened my coding discipline, version control knowledge, project management abilities, and understanding of collaborative software development practices.

Chapter 4

TECHNICAL TRAINING AND ASSIGNMENT

This chapter explains the complete development and implementation process of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project carried out during the Infosys Springboard Virtual Internship Program. It covers the technologies, frameworks, analytical techniques, dashboard development methods, AI-based forecasting systems, portfolio management features, and deployment processes used throughout the project development lifecycle. The main objective of the project was to build an intelligent cryptocurrency portfolio management platform capable of providing real-time market analysis, AI-generated trading signals, portfolio tracking, forecasting insights, and risk analysis through an interactive web-based dashboard. To achieve this, modern Python-based technologies and analytical libraries were integrated to ensure efficient data processing, smooth visualization, and an improved user experience.

The development process involved multiple stages such as requirement analysis, dashboard design, cryptocurrency market data processing, portfolio management integration, forecasting model implementation, AI-based analytical feature development, visualization design, testing, debugging, and cloud deployment. Special attention was given to creating a responsive, user-friendly, and data-driven application that could help users manage cryptocurrency investments and make better financial decisions. The project was mainly developed using Python along with the Streamlit framework for building interactive dashboards and frontend components.

Libraries such as Pandas, NumPy, and Plotly were used for financial analysis, numerical computations, market trend analysis, and interactive data visualization. AI-based forecasting and portfolio risk analysis modules were also implemented within the CRYPTOPORT application to provide predictive insights and intelligent investment recommendations based on cryptocurrency market behavior and trend analysis. These modules helped users understand possible future market movements, evaluate risks, and make better decisions using data-driven analytical insights. It covers dashboard integration techniques, cryptocurrency portfolio analysis methods, AI-assisted trading signal generation, forecasting implementation, and debugging activities, and performance throughout the development lifecycle of the CRYPTOPORT platform.



Fig 4.1: Training Program Flow

Fig 4.1 illustrates the complete workflow of the internship training and project development process followed during the Infosys Springboard Virtual Internship Program. The figure shows different stages including program initiation, technical training, skill development, project implementation, testing, deployment, review, and final assessment. It highlights how technical knowledge and practical skills were progressively applied during the development of the Crypto Portfolio Manager project.

4.1 Python Programming and Cryptocurrency Data Analysis

Python programming and cryptocurrency data analysis formed the core foundation of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project developed during the internship. Python was selected as the primary programming language because of its simplicity, flexibility, powerful analytical capabilities, and strong support for financial data processing and visualization. The project required functionalities such as real-time cryptocurrency market monitoring, portfolio analysis, forecasting, profit/loss evaluation, and portfolio risk assessment. Python provided an efficient and reliable environment for implementing these features through its wide range of data science, analytical, and visualization libraries.

Using Python, cryptocurrency market datasets were collected, processed, analyzed, and visualized to generate meaningful financial insights for users. The language also supported the implementation of AI-based forecasting models, portfolio

management systems, interactive dashboards, and analytical calculations required for intelligent investment analysis. Python's integration with libraries such as Pandas, NumPy, Plotly, and Streamlit further simplified the development process and improved application performance. These tools helped in managing financial datasets, performing numerical computations, generating forecasting insights, and creating interactive visual dashboards for real-time cryptocurrency analysis.

Data analysis played a major role in understanding cryptocurrency market trends, historical price movements, asset performance, and investment behavior. Libraries such as Pandas and NumPy were used to perform data cleaning, numerical computations, trend analysis, portfolio calculations, and volatility measurements. These analytical operations helped generate accurate portfolio insights and forecasting outputs. Python was also integrated with interactive visualization tools and the Streamlit framework to develop dynamic dashboards for displaying market trends, forecasting graphs, AI-based trading signals, portfolio performance, and risk analysis reports.

4.1.1 Core Python Concepts for Financial Applications

Python provides several important programming concepts that are highly useful for developing financial and analytical applications. During the development of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project, these concepts were widely used to implement cryptocurrency market analysis, portfolio tracking, forecasting systems, and portfolio risk evaluation features. Python variables and data types were used throughout the application to store cryptocurrency prices, investment values, portfolio details, profit/loss calculations, forecasting outputs, and analytical records efficiently. Conditional statements such as if, elif, and else were implemented to generate AI-based Buy, Hold, and Sell signals depending on market trends and forecasting results.

Loops such as for and while were used to process cryptocurrency datasets, analyze market trends, perform repetitive financial calculations, and dynamically generate portfolio analytics. Functions and modular programming techniques helped organize the application into reusable and manageable components for financial analysis, forecasting, portfolio management, and visualization modules. Different Python data structures such as lists, dictionaries, and arrays were used for storing cryptocurrency market data, portfolio information, forecasting results, and risk analysis

records. File handling and data processing techniques were also implemented to import, manage, and process real-time cryptocurrency datasets efficiently for analysis and dashboard integration.

Mathematical and numerical computation concepts played an important role in implementing profit/loss calculations, asset allocation analysis, portfolio return evaluation, volatility measurement, forecasting models, and risk assessment functionalities within the application. Python was further integrated with libraries such as Plotly and Streamlit to create interactive charts, portfolio dashboards, forecasting graphs, and analytical visualizations for better financial understanding. These core Python concepts formed the technical foundation of the CRYPTOPORT platform and helped in developing intelligent financial analysis, portfolio management, forecasting, and cryptocurrency investment monitoring functionalities efficiently.

4.1.2 Cryptocurrency Market Data Processing

Pandas and NumPy played an important role in processing and analyzing cryptocurrency market data within the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. These libraries provided efficient tools for handling large financial datasets, performing numerical computations, and generating analytical insights required for portfolio management and forecasting functionalities. Pandas was mainly used for collecting, organizing, cleaning, and managing cryptocurrency datasets obtained from different market sources. It helped structure both real-time and historical cryptocurrency data into tabular formats using DataFrames, which made financial analysis and data manipulation easier and more efficient.

NumPy was used for performing high-speed numerical and mathematical operations related to cryptocurrency analysis. It supported calculations such as profit/loss evaluation, portfolio return analysis, volatility measurement, forecasting computations, and portfolio risk analysis. The library improved computational efficiency and enabled fast processing of large numerical datasets within the application. By using Pandas and NumPy together, the project successfully implemented features such as cryptocurrency market trend analysis, real-time market monitoring, financial calculations, and forecasting analytics. These libraries also supported the generation of AI-based trading signals and risk assessment modules by providing accurate and efficient analytical computations. Pandas was mainly used for collecting, organizing, and cleaning datasets obtained from different sources.

The processed analytical data was further integrated into interactive dashboards and visualization systems using Plotly and Streamlit. This helped users better understand cryptocurrency market behavior, portfolio performance, and forecasting insights through graphical representation, real-time analytics, and interactive financial visualizations.

4.2 Interactive Dashboard Development with Streamlit

Streamlit played a vital role in the development of the interactive dashboard system for the Crypto Portfolio Manager project. The framework was used to create a responsive, user-friendly, and real-time cryptocurrency analytics platform capable of displaying portfolio information, market trends, forecasting insights, AI-based trading signals, and risk analysis reports. The dashboard was designed to provide users with a centralized platform for monitoring cryptocurrency investments and analyzing market behavior efficiently. Streamlit enabled the rapid development of interactive web interfaces directly using Python, eliminating the need for complex frontend technologies such as HTML, CSS, or JavaScript.

Various interactive components such as sidebars, dropdown menus, buttons, forms, charts, tables, and navigation panels were implemented using Streamlit. These features improved application usability and allowed users to access different analytical modules smoothly within the dashboard environment. The dashboard integrated real-time cryptocurrency market data, portfolio analytics, forecasting graphs, and risk analysis visualizations to provide meaningful financial insights.

Special attention was given to dashboard responsiveness, navigation flow, and user interaction design to ensure a professional and intuitive user experience. The dashboard was organized into multiple modules such as Market Dashboard, Portfolio Management, AI Trading Signals, Forecasting System, Risk Analysis, and AI Chat Assistant for efficient accessibility and functionality. The Streamlit-based dashboard significantly improved the visualization and presentation of financial analytics while simplifying the implementation of real-time cryptocurrency monitoring and AI-assisted portfolio management features within the CRYPTOPORT platform.

4.2.1 Setting Up and Configuring the Streamlit Environment

The initial stage of developing the Crypto Portfolio Manager project involved setting up and configuring the Streamlit development environment. Streamlit was selected as the primary framework because of its simplicity, flexibility, and ability to create

interactive analytical dashboards directly using Python. The development environment was configured by installing Python and required libraries such as Streamlit, Pandas, NumPy, Plotly, and other analytical packages necessary for cryptocurrency market analysis and portfolio management. Package management tools such as pip were used to install and manage dependencies efficiently.

A structured project directory was created to organize application modules, datasets, dashboard components, forecasting systems, portfolio analytics, and visualization files systematically. This modular project structure improved maintainability, debugging, and future scalability of the application. The Streamlit environment was configured to support interactive dashboard components such as sidebars, charts, forms, buttons, tables, and navigation systems. Integration with visualization libraries and analytical modules was also established during the setup process to enable real-time cryptocurrency data analysis and graphical representation.

Testing and configuration checks were performed to ensure proper installation of all required libraries and smooth execution of Streamlit applications. The local development server was then launched using Streamlit commands, enabling real-time testing and dashboard preview during application development. The configured development environment provided a stable and efficient foundation for implementing dashboard functionalities, portfolio management modules, forecasting systems, AI-based trading signals, and risk analysis features within the CRYPTOPORT platform.

4.2.2 Building Dashboard Components and User Interface

The development of dashboard components and user interface design was an important part of the Crypto Portfolio Manager project. Using Streamlit, an interactive and user-friendly dashboard was created to provide efficient access to cryptocurrency analytics, portfolio management tools, forecasting systems, and AI-based investment insights. Various dashboard components such as sidebars, navigation menus, buttons, forms, dropdown lists, tables, and interactive charts were implemented to improve application usability and navigation. These components allowed users to access different sections of the platform such as Market Dashboard, Portfolio Analysis, AI Trading Signals, Forecasting System, Risk Analysis, and AI Chat Assistant efficiently.

The user interface was designed with a focus on simplicity, responsiveness, and professional appearance. Streamlit layout features were used to organize dashboard elements systematically and present financial analytics in a visually

clear and understandable format. Interactive charts and graphical visualizations created using Plotly were integrated into the dashboard to display cryptocurrency market trends, portfolio distribution, forecasting analytics, and investment performance dynamically. Real-time updates and responsive UI elements improved the overall user experience and platform interaction. Custom styling and layout organization techniques were also applied to enhance dashboard appearance, improve readability, and provide smooth user navigation throughout the application. The dashboard structure ensured efficient presentation of complex financial data and AI-generated insights in an accessible format.

4.3 Portfolio Management and Risk Analysis System

The Portfolio Management and Risk Analysis System was one of the major modules developed in the Crypto Portfolio Manager project. This system was designed to help users efficiently manage cryptocurrency investments, monitor portfolio performance, evaluate financial risks, and make informed investment decisions using analytical insights and real-time market data. The portfolio management module allowed users to track cryptocurrency holdings, investment values, profit/loss status, asset allocation, and portfolio diversification dynamically. Using Python along with analytical libraries such as Pandas and NumPy, the system performed various financial calculations related to portfolio performance and investment analytics.

Real-time cryptocurrency market prices were integrated into the system to continuously update portfolio values and market performance indicators. The module dynamically calculated metrics such as current investment value, total portfolio balance, percentage returns, and individual asset contribution to provide accurate portfolio insights. The risk analysis component evaluated investment stability and market exposure by analyzing asset volatility, portfolio diversification, and cryptocurrency market fluctuations. Risk scores and portfolio health indicators were generated to help users identify high-risk investments and maintain balanced portfolio strategies.

Interactive charts and analytical visualizations were implemented using Plotly and integrated into the Streamlit dashboard to display portfolio distribution, profit/loss analysis, risk assessment graphs, and investment performance metrics in a visually understandable format. The Portfolio Management and Risk Analysis System significantly improved the intelligence and usability of the CRYPTOPORT platform

by combining real-time financial analytics, portfolio monitoring, risk evaluation, and graphical reporting into an interactive cryptocurrency management environment.

4.3.1 Portfolio Tracking and Asset Management

Portfolio Tracking and Asset Management were important functionalities implemented in the Crypto Portfolio Manager project to help users efficiently manage cryptocurrency investments and monitor financial performance in real time. The system was designed to provide a centralized platform for tracking multiple cryptocurrency assets, investment values, and portfolio growth dynamically. Using Python along with analytical libraries such as Pandas and NumPy, the application continuously processed cryptocurrency market data and portfolio records to calculate investment metrics and portfolio performance indicators.

The portfolio tracking module allowed users to view details such as cryptocurrency names, quantity of assets held, buy prices, current market prices, investment value, and profit/loss status. Real-time market data integration ensured that portfolio analytics remained updated according to current cryptocurrency price movements. Asset management features were implemented to organize holdings, analyze asset allocation, and monitor investment diversification within the portfolio.

The system also calculated overall portfolio value, percentage returns, and individual asset contribution to provide meaningful financial insights. Interactive tables, portfolio distribution charts, and analytical dashboards were integrated into the Streamlit interface to improve visualization and user understanding of portfolio performance. The implementation of Portfolio Tracking and Asset Management significantly improved the functionality of the CRYPTOPORT platform by providing efficient investment monitoring, real-time analytics, and intelligent portfolio management capabilities.

4.3.2 Profit/Loss Analysis and Investment Monitoring

Profit/Loss Analysis and Investment Monitoring were essential components of the Crypto Portfolio Manager project. These functionalities were developed to help users analyze cryptocurrency investment performance, monitor portfolio growth, and evaluate financial gains or losses in real time. Using Python along with analytical libraries such as Pandas and NumPy, the application performed automated financial calculations based on cryptocurrency market prices and user investment records. The system continuously monitored current cryptocurrency prices and compared them with

initial investment values to calculate profit or loss for each asset within the portfolio. Metrics such as total investment amount, current portfolio value, percentage returns, net profit/loss, and individual asset performance were dynamically generated and displayed through the dashboard.

Investment monitoring features enabled users to track portfolio performance in real time and analyze changes in cryptocurrency market conditions. The system provided updated analytical insights regarding asset growth, declining investments, and portfolio trends to support better financial decision-making. Interactive charts and graphical visualizations were integrated using Plotly to display profit/loss analysis, investment growth trends, and portfolio performance comparisons in a visually understandable format. These visual analytics improved user understanding of market behavior and financial performance.

4.4 AI-Based Forecasting and Trading Signal Implementation

AI-Based Forecasting and Trading Signal Implementation was one of the most advanced and intelligent modules developed in the Crypto Portfolio Manager project. This module was designed to analyze cryptocurrency market trends, predict future market behavior, and generate investment recommendations to assist users in making informed financial decisions. Using Python and analytical libraries such as Pandas and NumPy, historical cryptocurrency market data was processed and analyzed to identify price movement patterns, market trends, volatility behavior, and investment opportunities.

The forecasting system utilized predictive analysis techniques to estimate future cryptocurrency price trends based on historical market performance and statistical calculations. These forecasting outputs helped users understand possible future market conditions and supported better portfolio planning strategies. An AI-based trading signal system was implemented to generate Buy, Hold, and Sell recommendations dynamically according to market conditions, trend analysis, and forecasting results. The system continuously monitored market fluctuations and portfolio behavior to provide real-time analytical insights and trading suggestions.

Interactive forecasting graphs, market trend charts, and analytical dashboards were developed using Plotly and integrated into the Streamlit dashboard. These visualizations improved user understanding of cryptocurrency behavior and forecasting results through graphical representation. The AI-based forecasting and trading signal

module significantly enhanced the intelligence and analytical capability of the CRYPTOPORT platform by combining predictive analysis.

4.4.1 Cryptocurrency Market Forecasting Techniques

Cryptocurrency Market Forecasting Techniques were implemented in the CRYPTOPORT – AI Powered Crypto Portfolio Manager project to analyze historical market behavior and predict possible future cryptocurrency price movements. These forecasting techniques helped users understand market trends, identify investment opportunities, and make better financial decisions based on analytical insights and predictive market analysis. Using Python along with analytical libraries such as Pandas and NumPy, historical cryptocurrency market data including price trends, market fluctuations, and trading patterns was processed for predictive analysis. Time-series analysis techniques were applied to study historical cryptocurrency prices and identify market movement patterns across different time periods.

Trend analysis methods were used to identify upward and downward market behavior, while statistical calculations helped evaluate market volatility and investment stability. Moving averages and trend indicators were implemented to reduce short-term market fluctuations and analyze long-term cryptocurrency performance more effectively. These techniques improved forecasting accuracy and generated more reliable market trend insights. The forecasting system also utilized analytical calculations and predictive modeling techniques to estimate future cryptocurrency price movements and generate forecasting outputs for selected digital assets. These predictive results were visualized through interactive charts, forecasting graphs, and analytical dashboards integrated into the Streamlit-based application interface.

Real-time cryptocurrency market data integration further improved forecasting efficiency by continuously updating analytical models according to current market conditions. This enabled the application to provide dynamic forecasting insights and support real-time investment decision-making within the dashboard environment. Forecasting visualizations and analytical dashboards helped users better understand future market possibilities through graphical representation and trend analysis. Interactive forecasting charts improved user engagement and simplified complex financial analysis by presenting predictive insights in an organized and user-friendly manner. The Cryptocurrency Market Forecasting Techniques module significantly

enhanced the analytical intelligence and decision-support capabilities of the CRYPTOPORT platform by combining statistical analysis, predictive modeling, real-time data processing, and interactive visualization within a unified cryptocurrency forecasting system.

4.4.2 AI-Based Buy, Hold, and Sell Signal Generation

Pandas and NumPy played an important role in processing and analyzing cryptocurrency market data within the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. These libraries provided efficient tools for handling large financial datasets, performing numerical computations, and generating analytical insights required for portfolio management and forecasting functionalities. Pandas was mainly used for collecting, organizing, cleaning, and managing cryptocurrency datasets obtained from different market sources. It helped structure both real-time and historical cryptocurrency data into tabular formats using DataFrames, which made financial analysis and data manipulation easier and more efficient.

The library was also used for filtering market records, sorting datasets, handling missing values, and preparing data for forecasting and visualization processes. NumPy was used for performing high-speed numerical and mathematical operations related to cryptocurrency analysis. It supported calculations such as profit/loss evaluation, portfolio return analysis, volatility measurement, forecasting computations, and portfolio risk analysis. The library improved efficiency and enabled fast processing of large numerical datasets within the application. By using Pandas and NumPy together, the project successfully implemented features such as cryptocurrency market trend analysis, market monitoring, portfolio performance evaluation, financial calculations, and forecasting analytics.

These libraries also supported the implementation of AI-based trading signal generation and portfolio risk assessment modules by providing accurate and efficient analytical computations. The processed financial and analytical data was further integrated into interactive dashboards and visualization systems using Plotly and Streamlit. This integration helped users better understand cryptocurrency market behavior, portfolio performance, forecasting insights, and investment trends through graphical representation, real-time analytics, and interactive financial visualizations. The combination of analytical processing and visualization technologies significantly enhanced the responsiveness, intelligence, and user experience of the platform.

4.5 Data Visualization and Interactive Analytics

Data Visualization and Interactive Analytics were essential components of the Crypto Portfolio Manager project. These functionalities were implemented to present complex cryptocurrency market data, portfolio analytics, forecasting results, and risk assessment reports in a visually understandable and interactive format for users. Using Plotly along with Streamlit, various graphical dashboards and analytical charts were developed to improve user understanding of cryptocurrency market behavior and portfolio performance. Interactive visualizations significantly enhanced the overall usability and analytical capability of the application.

The system displayed real-time cryptocurrency price trends, portfolio distribution, forecasting graphs, investment growth analysis, volatility charts, and profit/loss reports through dynamic graphical interfaces. These visualizations allowed users to monitor market movements and financial performance efficiently. Interactive analytics features such as chart zooming, filtering, hover information, and dynamic updates were integrated into the dashboard to improve user interaction and provide detailed analytical insights. Users could analyze different cryptocurrencies, compare portfolio assets, and monitor investment trends in real time.

Data visualization also played an important role in presenting AI-based forecasting outputs and trading signals. Predictive market trends, Buy/Hold/Sell recommendations, and risk analysis metrics were displayed using analytical charts and dashboard reports to support data-driven investment decisions. The integration of real-time analytics and interactive visual dashboards significantly improved the professional quality and intelligence of the CRYPTOPORT platform by simplifying financial analysis and enhancing user engagement with cryptocurrency market data.

4.5.1 Interactive Chart Development Using Plotly

Plotly was extensively used in the Crypto Portfolio Manager project to develop interactive charts and analytical visualizations for cryptocurrency market analysis and portfolio monitoring. Plotly provided advanced graphical capabilities that helped present complex financial data in a visually understandable and user-friendly format. Various interactive charts such as line graphs, bar charts, pie charts, forecasting visuals, and portfolio distribution graphs were implemented within the Streamlit dashboard. These visualizations enabled users to monitor cryptocurrency price trends, portfolio performance, investment growth, and market fluctuations dynamically.

Interactive features such as zooming, hovering, filtering, and real-time chart updates were integrated into the dashboard to improve user interaction and make financial analysis more engaging and user-friendly. These features allowed users to explore detailed cryptocurrency information, compare market trends, and monitor portfolio performance more effectively through interactive graphical dashboards. Plotly was also used to visualize AI-based forecasting results, trading signals, and portfolio risk analysis metrics within the application.

Predictive trend graphs, forecasting charts, and portfolio analytics dashboards helped users better understand cryptocurrency market behavior and make more informed investment decisions based on analytical insights. The integration of Plotly significantly improved the visual appearance, responsiveness, and analytical capabilities of the CRYPTOPORT platform. It enabled the development of professional interactive dashboards and real-time financial visualizations that enhanced overall user experience and simplified complex cryptocurrency data analysis.

4.5.2 Cryptocurrency Trend Visualization

Cryptocurrency Trend Visualization was implemented in the Crypto Portfolio Manager project to help users analyze cryptocurrency market behavior and monitor price movements through graphical representations and analytical dashboards. Using Plotly along with Pandas, real-time and historical cryptocurrency market data was processed and visualized through interactive trend charts and analytical graphs. The visualization system displayed market movement patterns, price fluctuations, asset performance trends, and historical market behavior for different cryptocurrencies. Line charts and analytical graphs were used to represent upward and downward market movements over different time intervals.

Real-time market data integration enabled the trend visualization system to continuously update charts according to changing cryptocurrency prices and market conditions. This allowed users to monitor live market trends and evaluate investment opportunities efficiently. Trend visualization dashboards also supported forecasting analysis and AI-based trading signal systems by presenting predictive market movements and investment insights graphically. These visualizations improved user understanding of market conditions.

Chapter 5

INTERNSHIP PROJECT CONTRIBUTION

This chapter describes the practical contributions and development work carried out during the Infosys Springboard Virtual Internship Program for the project “Crypto Portfolio Manager.” The chapter highlights the implementation of major project modules, dashboard development activities, analytical systems, forecasting functionalities, portfolio management features, and AI-based cryptocurrency analysis developed throughout the internship period. The internship project focused on building a professional web-based cryptocurrency portfolio management platform capable of providing real-time market monitoring, portfolio tracking, forecasting analysis, AI-generated trading signals, and risk evaluation through an interactive dashboard environment. The project was designed to assist users in managing cryptocurrency investments and making informed financial decisions using analytical insights and predictive technologies.

During the internship, contributions were made in multiple areas including Python programming, cryptocurrency data analysis, dashboard development using Streamlit, interactive visualization implementation using Plotly, portfolio analytics, forecasting systems, and AI-based trading signal generation. Significant work was also carried out in integrating real-time market data, developing risk analysis modules, and improving user experience through responsive dashboard design. The project additionally involved implementation of performance tracking systems, forecasting dashboards, volatility analysis, and AI chat assistance modules. Real-time cryptocurrency analytics and interactive graphical reporting systems were integrated to enhance application usability and provide meaningful financial insights to users.

Cloud deployment and version control management were also important parts of the contribution process. The final application was successfully deployed using Streamlit Community Cloud, enabling online accessibility and real-time interaction through web browsers. Git and GitHub were used for project management, source code tracking, and version control throughout the development lifecycle. The internship project contribution provided valuable practical exposure to software development, financial analytics, AI-assisted systems, dashboard engineering, and modern web application implementation while significantly improving technical and professional skills in the field of fintech and cryptocurrency portfolio management.

5.1 Project Understanding and Cryptocurrency Market Research

The initial phase of the internship project involved understanding the objectives, workflow, and functional requirements of the **Crypto Portfolio Manager** system. The primary goal of the project was to develop an intelligent cryptocurrency portfolio management platform capable of providing real-time market monitoring, portfolio analysis, forecasting insights, AI-based trading signals, and risk evaluation through an interactive dashboard environment. The internship project focused on building a professional web-based cryptocurrency portfolio management platform capable of providing real-time market monitoring, portfolio tracking, forecasting analysis, AI-generated trading signals, and risk evaluation through an interactive dashboard environment. The project was designed to assist users in managing cryptocurrency investments and making informed financial decisions using analytical insights and predictive technologies.

During the internship, contributions were made in multiple areas including Python programming, cryptocurrency data analysis, dashboard development using Streamlit, interactive visualization implementation using Plotly, portfolio analytics, forecasting systems, and AI-based trading signal generation. Significant work was also carried out in integrating real-time market data, developing risk analysis modules, and improving user experience through responsive dashboard design. The project additionally involved implementation of portfolio performance tracking systems, asset management functionalities, forecasting dashboards, volatility analysis, and AI chat assistance modules. Real-time cryptocurrency analytics and interactive graphical reporting systems were integrated to enhance application usability and provide meaningful financial insights to users.

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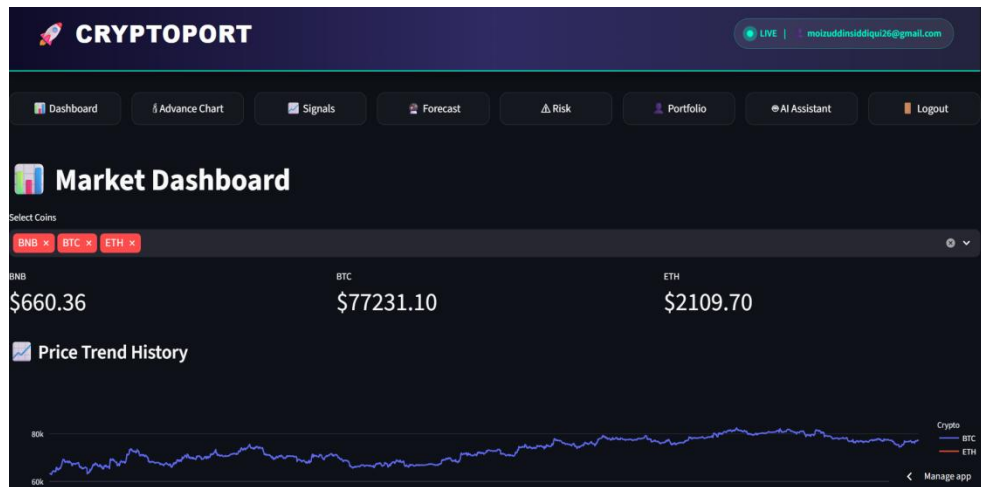


Fig 5.1: Application Screenshot Look Alike

Fig 5.1 shows the main dashboard interface of the **Crypto Portfolio Manager** application developed during the internship. The dashboard provides a centralized platform for monitoring real-time cryptocurrency market data, portfolio analytics, forecasting insights, and AI-based financial analysis. The interface was developed using Streamlit and designed with a modern, responsive, and user-friendly layout to improve accessibility and interaction.

5.2 AI-Based Trading Signal Generation and Forecasting

The AI-Based Trading Signal Generation and Forecasting module was one of the most advanced functionalities implemented in the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. The primary objective of this system was to analyze cryptocurrency market behavior, predict future market trends, and generate intelligent Buy, Hold, and Sell recommendations for users based on analytical calculations and forecasting techniques. Using Python along with analytical libraries such as Pandas and NumPy, the application continuously processed real-time and historical cryptocurrency datasets to identify market patterns and investment trends.

The forecasting system analyzed market indicators such as price fluctuations, historical movement patterns, volatility, trend direction, and trading performance to estimate future cryptocurrency behavior. Mathematical models and statistical analysis techniques were applied to improve forecasting accuracy and generate predictive insights for portfolio management. Using Python along with analytical libraries such as Pandas and NumPy. The AI-based trading signal system evaluated processed market data and generated intelligent recommendations such as Buy, Hold, or Sell depending on current market conditions and analytical outputs.

Real-time forecasting outputs and AI-generated signals were integrated into the interactive dashboard using Streamlit and Plotly visualizations. Graphical forecasting charts, market trend indicators, and analytical reports were displayed dynamically to provide clear and understandable financial insights for users. Additional risk evaluation and volatility analysis were also integrated into the module to improve the reliability of trading recommendations and support investment strategies. The implementation of AI-Based Trading Signal Generation and Forecasting significantly enhanced the intelligence, automation, and analytical capabilities of the platform by providing predictive financial insights and intelligent investment guidance.

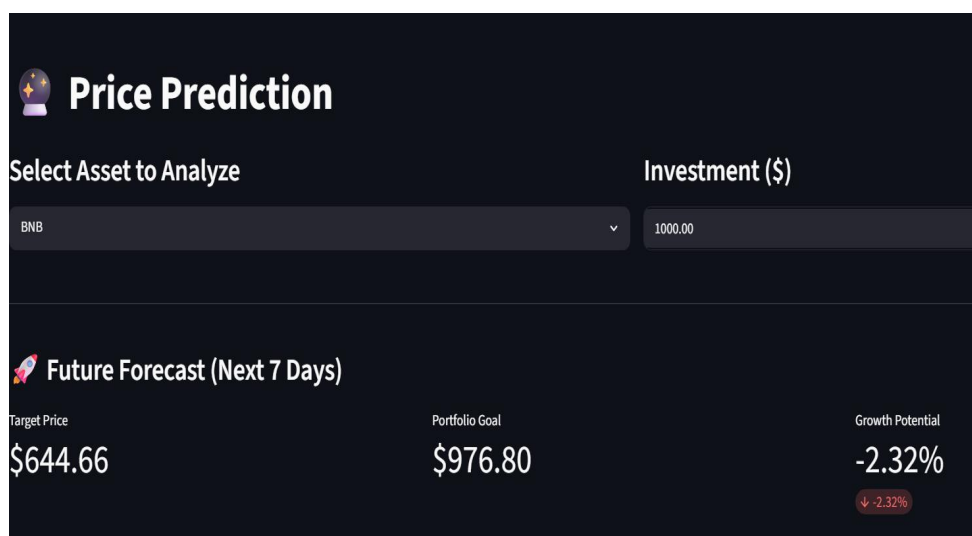


Fig 5.2: AI-Based Forecasting System

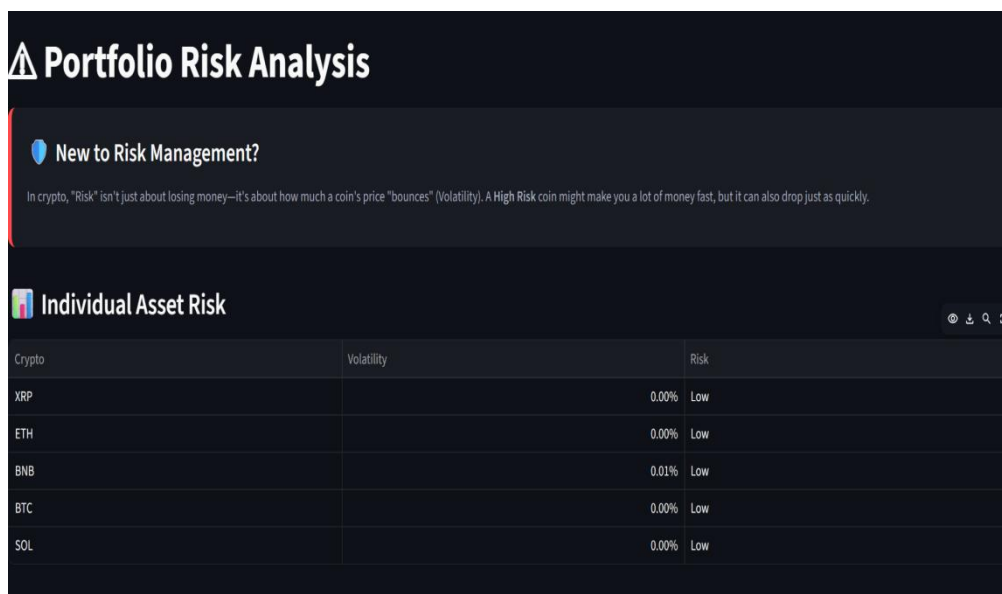
Fig 5.5 shows the AI-Based Trading Forecasting System developed in the CRYPTOPORT application. The dashboard analyzes cryptocurrency market trends and predicts future asset performance using forecasting techniques and analytical models. portfolio growth estimation, and future investment potential through interactive financial analytics and real-time forecasting visualization.

5.3 Portfolio Risk Analysis and Volatility Evaluation

Portfolio Risk Analysis and Volatility Evaluation were important analytical modules developed during the implementation of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. The objective of this system was to help users identify investment risks, analyze cryptocurrency market volatility, and maintain balanced portfolio strategies using real-time analytical insights. Using Python along with Pandas and NumPy, the application continuously processed cryptocurrency market data and portfolio records to calculate risk indicators and investment stability metrics.

The system analyzed factors such as price fluctuations, asset volatility, historical market trends, portfolio diversification, and investment performance to evaluate portfolio risk levels. Volatility calculations helped determine how rapidly cryptocurrency prices changed over time and assisted users in understanding potential investment exposure. Portfolio risk scores and stability indicators were dynamically generated and displayed within the dashboard environment. Interactive charts and analytical visualizations were integrated using Plotly to present volatility analysis, portfolio diversification, and portfolio risk evaluation reports in a visually understandable format. These visualizations helped users analyze investment behavior, compare asset stability, and monitor portfolio performance more effectively through graphical dashboards.

The implementation of Portfolio Risk Analysis and Volatility Evaluation significantly improved the financial intelligence and analytical capabilities of the CRYPTOPORT platform by helping users make safer and more balanced cryptocurrency investment decisions. Real-time market monitoring and analytical calculations further enhanced the accuracy of risk evaluation and investment analysis within the application. The portfolio risk dashboard simplified complex financial analytics by presenting risk metrics and investment insights through organized visual representations. This improved user understanding of cryptocurrency market uncertainty and supported more informed and data-driven portfolio management decisions.



Crypto	Volatility	Risk
XRP	0.00%	Low
ETH	0.00%	Low
BNB	0.01%	Low
BTC	0.00%	Low
SOL	0.00%	Low

Fig 5.3: Portfolio Risk Analysis

Fig 5.6 shows the Portfolio Risk Analysis and Volatility Evaluation System implemented in the CRYPTOPORT application. The dashboard analyzes cryptocurrency asset volatility and evaluates portfolio risk levels based on market fluctuations and investment behavior. It provides individual asset risk assessment, volatility measurement, and financial risk insights to help users understand portfolio stability and make informed investment decisions.

5.4 Styling Interactive Charts and Financial Data Visualization

Interactive charts and financial data visualization played a major role in improving the usability and analytical presentation of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. The visualization system was designed to represent complex cryptocurrency market data and portfolio analytics through dynamic graphical dashboards and real-time financial charts. Using Plotly along with Streamlit integration, multiple interactive visualizations such as line charts, bar graphs, pie charts, forecasting plots, and portfolio allocation diagrams were implemented within the application dashboard.

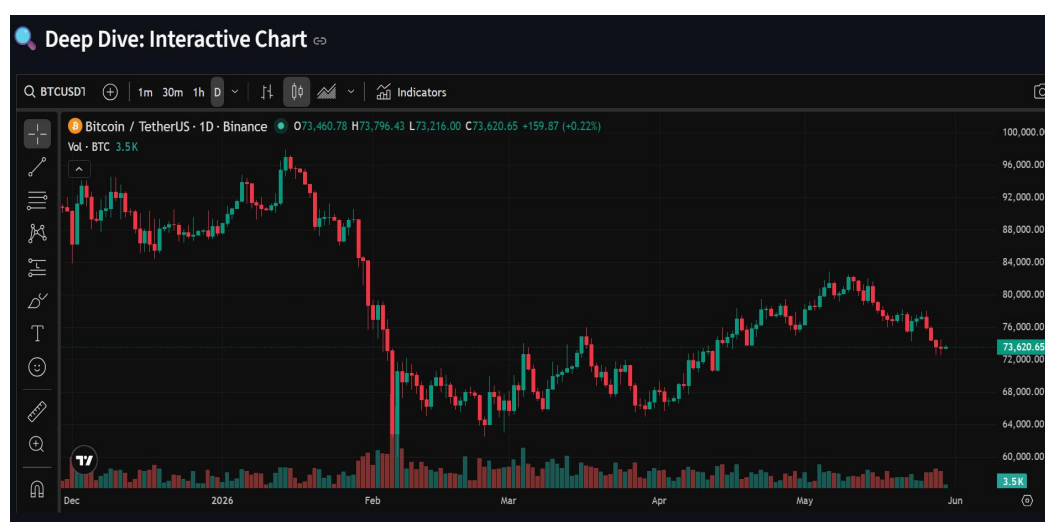


Fig 5.4: Interactive Charts and Financial Data

Fig 5.7 shows the Interactive Charts and Financial Data Visualization Dashboard implemented in the CRYPTOPORT application. The system provides advanced cryptocurrency market analysis through interactive candlestick charts, trading indicators, market trends, and real-time graphical visualization tools. The integration of interactive visualization tools significantly enhanced the analytical capabilities and professional appearance of the platform. These visualizations also supported better investment decision-making by helping users identify market trends, trading patterns, and potential forecasting opportunities more effectively.

The system displayed cryptocurrency price trends, market fluctuations, forecasting results, portfolio performance, profit/loss analysis, and risk evaluation reports using graphical representation techniques. Interactive chart functionalities such as zooming, hovering, filtering, and dynamic updates improved user interaction and financial understanding. Real-time data visualization enabled users to monitor cryptocurrency markets continuously and interpret financial insights more effectively. Graphical dashboards simplified analytical interpretation and enhanced the overall user experience within the CRYPTOPORT platform. The implementation of interactive charts and financial visualization significantly improved the analytical reporting and dashboard intelligence of the application by transforming raw financial data into meaningful visual insights.

5.5 Portfolio Management and Investment Tracking System

The Portfolio Management and Investment Tracking System was one of the core modules developed during the implementation of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. This system was designed to help users efficiently manage cryptocurrency investments, monitor portfolio performance, and track financial growth using real-time analytical insights and interactive dashboard features. Using Python along with Pandas and NumPy, the system continuously processed cryptocurrency market data and user investment information to generate accurate portfolio analytics and financial reports. The portfolio management module enabled users to manage multiple cryptocurrency assets within a single dashboard environment.

Features such as asset tracking, portfolio valuation, profit/loss monitoring, investment distribution, and performance analysis were implemented to improve investment management efficiency. The investment tracking system dynamically calculated important financial metrics including current portfolio value, total investment amount, percentage returns, individual asset contribution, and overall portfolio performance. Real-time cryptocurrency market prices were integrated into the application to ensure continuously updated investment analytics. Interactive tables, portfolio allocation charts, and analytical dashboards were developed using Plotly and integrated into the Streamlit interface to improve data visualization and financial understanding. Users could monitor cryptocurrency holdings, compare asset performance, and evaluate investment growth and real-time insights.

Portfolio analysis functionalities such as asset allocation monitoring, diversification tracking, and investment performance comparison were also implemented to support better financial decision-making and portfolio optimization strategies. The Portfolio Management and Investment Tracking System significantly enhanced the analytical intelligence and usability of the CRYPTOPORT platform by providing efficient cryptocurrency investment management, real-time financial monitoring, and interactive portfolio analytics capabilities.

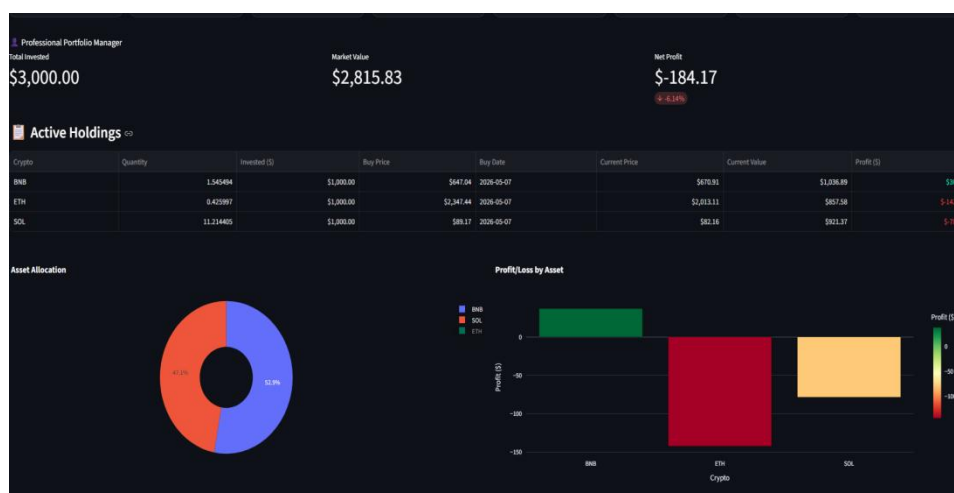


Fig 5.5: Investment Tracking System

Fig 5.4 shows the Portfolio Management and Investment Tracking Dashboard developed in the CRYPTOPORT application. The dashboard provides real-time portfolio analytics including total investment, market value, net profit/loss, active cryptocurrency holdings, asset allocation visualization, and profit/loss comparison for different crypto assets. Interactive charts and graphical analysis help users monitor portfolio performance, evaluate investment growth, and analyze asset distribution efficiently.

5.6 Application Optimization, Testing, and Debugging

Application Optimization, Testing, and Debugging were important stages carried out during the final development phase of the CRYPTOPORT – AI Powered Crypto Portfolio Manager project. These activities were mainly focused on improving system performance, ensuring application reliability, and enhancing the overall user experience of the platform. Different testing methods were applied to verify the proper functioning of various modules such as the dashboard system, portfolio management features, forecasting modules, AI-generated trading signals, and interactive visualization components.

Functional testing helped identify coding errors, performance issues, and system inconsistencies during the development process, ensuring that each feature worked accurately and efficiently. Debugging techniques and Python error-handling mechanisms were implemented to resolve coding issues, improve application stability, and optimize dashboard responsiveness. Several performance optimization activities were also carried out to reduce loading time, improve data processing efficiency, and ensure smooth execution of real-time analytical operations within the platform. Special attention was given to improving dashboard responsiveness, chart rendering speed, and smooth navigation between different modules of the application.

Real-time cryptocurrency data handling and interactive visualizations were optimized to maintain stable performance even during continuous analytical updates. The testing and optimization process significantly improved the quality, efficiency, stability, and reliability of the CRYPTOPORT platform before deployment.

5.7 Cloud Deployment, Version Control, and Final Enhancements

The final stage of the project involved cloud deployment, version control management, and implementation of final enhancements for the CRYPTOPORT – AI Powered Crypto Portfolio Manager application. The main objective of this phase was to deploy the completed application online, maintain proper source code organization, and improve the overall functionality and presentation of the platform. The application was deployed using Streamlit Community Cloud, which enabled users to access the dashboard directly through a web browser without requiring local installation. Cloud deployment improved the accessibility, scalability, and real-time availability of the application while making it easier to demonstrate and use the platform online.

Git and GitHub were used throughout the project for source code management, version tracking, repository maintenance, and project backup. Several final enhancements such as dashboard optimization, user interface improvements, responsive layouts, visualization refinements, and performance tuning were also implemented to improve overall application quality and user experience. The successful completion of deployment and final optimization activities marked the completion of the project and demonstrated the practical implementation of cryptocurrency analytics, AI-based forecasting, portfolio management, cloud deployment, and interactive dashboard development technologies within a real-world fintech application.

CONCLUSION

The Infosys Springboard Virtual Internship Program provided valuable practical exposure to real-world software development, financial analytics, and modern web application technologies. Throughout the internship, significant technical and professional skills were developed while working on the project “CRYPTOPORT – AI Powered Crypto Portfolio Manager.” The internship experience helped bridge the gap between academic learning and industrial implementation by providing hands-on experience in developing a complete AI-powered cryptocurrency analytics platform.

The project successfully integrated various technologies such as Python, Streamlit, Pandas, NumPy, Plotly, Git, and GitHub to create an interactive and intelligent cryptocurrency portfolio management system. Multiple modules including real-time market analysis, portfolio management, AI-based trading signals, forecasting systems, risk analysis, interactive dashboards, and AI chatbot integration were implemented successfully during the development process.

The internship provided practical understanding of software development methodologies, financial data analysis, dashboard design, cloud deployment, and version control systems. Exposure to real-time cryptocurrency analytics and AI-assisted forecasting techniques significantly improved problem-solving ability, analytical thinking, debugging skills, and application development knowledge.

The CRYPTOPORT application demonstrated how modern technologies and data-driven analytics can be integrated to support cryptocurrency investment management and financial decision-making. The successful completion of the project enhanced confidence in handling real-world software projects and strengthened technical competencies in fintech application development and interactive dashboard implementation.

Overall, the internship experience at Infosys Springboard was highly beneficial in improving technical expertise, professional skills, project management capability, and practical industry exposure. The knowledge and experience gained through this internship will be valuable for future opportunities in software development, artificial intelligence, data analytics, and financial technology domains.

REFERENCES

- [1] D.,G. McKinney, *Python for Data Analysis*, 3rd ed., O'Reilly Media, 2022.
- [2] W. McKinney, "Pandas Documentation." [Online]. Available: <https://pandas.pydata.org/docs>
- [3] NumPy Developers, "NumPy Documentation." [Online]. Available: <https://numpy.org/doc>
- [4] Plotly Technologies Inc., "Plotly Python Graphing Library Documentation." [Online]. Available: <https://plotly.com/python>
- [5] Streamlit Inc., "Streamlit Documentation." [Online]. Available: <https://docs.streamlit.io>
- [6] Python Software Foundation, "Python Documentation." [Online]. Available: <https://docs.python.org/3>
- [7] Git SCM, *Pro Git Book*. [Online]. Available: <https://git-scm.com/book/en/v2>
- [8] GitHub, "GitHub Documentation." [Online]. Available: <https://docs.github.com>
- [9] Binance Academy, "Cryptocurrency Trading and Blockchain Fundamentals." [Online]. Available: <https://academy.binance.com>
- [10] CoinMarketCap, "Cryptocurrency Market Capitalizations." [Online]. Available: <https://coinmarketcap.com>
- [11] Yahoo Finance, "Cryptocurrency Market Data." [Online]. Available: <https://finance.yahoo.com>
- [12] Mozilla Developer Network, "Python and Web Development Resources." [Online]. Available: <https://developer.mozilla.org>
- [13] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.
- [14] J. Brownlee, *Machine Learning Mastery with Python*, Machine Learning Mastery, 2016.
- [15] Infosys Springboard, "Infosys Springboard Virtual Internship Program." [Online]. Available: <https://infyspringboard.onwingspan.com>